About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE DA 2025

A. Formulae

1.1.1 Indicator Summation for Thresholding

$$N = \sum_{i=1}^{M} \sum_{j=1}^{N} \mathbb{I}\left(U_{ij}\right) \tag{1.1.1.1}$$

Solution: The total count of elements in a matrix $\mathbf{U}$ satisfying a threshold c is obtained by summing the indicator function $\mathbb{I}$. The values of $\mathbb{I}$ operate as a step function defined as:

$$\mathbb{I}\left(U_{ij}\right) = \begin{cases} 1, & U_{ij} \leq c \\ 0, & U_{ij} > c \end{cases} \tag{1.1.1.2}$$

1.1.2 Geometric Scaling under Dilation

$$A' = (1+\alpha)\, A \tag{1.1.2.1}$$

Solution: For a rectangle with length L and width W, Area $A = LW$. If W is scaled by a factor of $(1+\alpha)$:

$$W' = (1+\alpha)\, W \tag{1.1.2.2}$$

Substituting (1.1.2.2) into the area equation:

$$A' = LW' = L\left[(1+\alpha)\, W\right] = (1+\alpha)\, LW = (1+\alpha)\, A \tag{1.1.2.3}$$

yielding (1.1.2.1).

1.1.3

$$x_1 < c < x_2 \quad \text{for} \quad f\left(x_1\right) = f\left(x_2\right) \tag{1.1.3.1}$$

Solution: If $f\left(x\right)$ strictly increases on $[0, c]$ and strictly decreases on $[c, \infty)$, each branch is individually injective (one-to-one). If two distinct points x_1 and x_2 yield the same function value, they cannot lie on the same monotonic branch, meaning the peak c must lie strictly between them.

1.1.4 Exponent Laws

$$\frac{a^u}{a^v} = a^{u-v}, \quad \left(a^x\right)^y = a^{xy} \tag{1.1.4.1}$$

1.1.5 Conditional Expectation and Law of Total Expectation

$$E[X] = \sum_k k p_X(k) \tag{1.1.5.1}$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \tag{1.1.5.2}$$

$$E[X \mid Y = y] = \sum_k k p_{X|Y}(k \mid y) \tag{1.1.5.3}$$

$$E[E[X \mid Y]] = \sum_y E[X \mid Y = y] p_Y(y) \tag{1.1.5.4}$$

1.1.6 Computational Complexity of Back-Substitution

For an $n \times n$ upper triangular matrix system $\mathbf{U}\mathbf{x} = \mathbf{b}$, variables are obtained sequentially:

$$x_i = \frac{1}{U_{ii}} \left(b_i - \sum_{j=i+1}^{n} U_{ij} x_j \right) \tag{1.1.6.1}$$

1.1.7 Eigenvalues of Constant Row-Sum Matrices and System Solvability

$$\mathbf{A}\mathbf{1} = R \cdot \mathbf{1}, \quad (\mathbf{A} - R\mathbf{I})\,\mathbf{1} = 0 \cdot \mathbf{1} \tag{1.1.7.1}$$

Solution: Let $\mathbf{a}_i^\top$ denote the i-th row of $\mathbf{A} \in \mathbb{R}^{n \times n}$. Because the sum of the elements in each row is R, multiplying the row by the all-ones column vector $\mathbf{1} = \begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix}^\top$ yields:

$$\mathbf{a}_i^\top \mathbf{1} = R, \quad \forall i \in \{1, 2, \dots, n\} \tag{1.1.7.2}$$

Grouping these row equations together to reconstruct the full matrix $\mathbf{A}$ gives:

$$\mathbf{A}\mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \\ \mathbf{a}_2^\top \\ \vdots \\ \mathbf{a}_n^\top \end{pmatrix} \mathbf{1} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{1} \\ \mathbf{a}_2^\top \mathbf{1} \\ \vdots \\ \mathbf{a}_n^\top \mathbf{1} \end{pmatrix} = \begin{pmatrix} R \\ R \\ \vdots \\ R \end{pmatrix} = R \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = R \cdot \mathbf{1} \tag{1.1.7.3}$$

This demonstrates that the vector $\mathbf{1}$ is an eigenvector of $\mathbf{A}$ associated with the eigenvalue $\lambda = R$.

To find the eigenvalue of $\mathbf{A} - R\mathbf{I}$ corresponding to the same eigenvector $\mathbf{1}$:

$$(\mathbf{A} - R\mathbf{I})\,\mathbf{1} = \mathbf{A}\mathbf{1} - (R\mathbf{I})\,\mathbf{1} = R\mathbf{1} - R\mathbf{1} = \mathbf{0} = 0 \cdot \mathbf{1} \tag{1.1.7.4}$$

Thus, $\mathbf{A} - R\mathbf{I}$ possesses an eigenvalue of 0, meaning it is a singular matrix.

Number of Solutions for Linear Systems:

i) For the system $(\mathbf{A} - R\mathbf{I})\,\mathbf{x} = \mathbf{b}$: Because $\mathbf{A} - R\mathbf{I}$ is singular (rank $< n$), the system will have infinitely many solutions if $\mathbf{b}$ is in the column space of $\mathbf{A} - R\mathbf{I}$, or no solution if it is not.

ii) For the system $\mathbf{A}\mathbf{x} = \mathbf{b}$: The number of solutions depends on R and the remaining unknown eigenvalues of $\mathbf{A}$. If $R \neq 0$ and all other eigenvalues are non-zero

($\left|\mathbf{A}\right| \neq 0$), the system has a unique solution. If $R = 0$ or 0 happens to be one of the other eigenvalues of $\mathbf{A}$, the matrix is singular, resulting in either no solution or infinitely many solutions.

1.1.8 Higher-Order Derivatives of Hyperbolic Sine Let $f(x) = \sinh(x)$.

$$f'(x) = \cosh(x) \tag{1.1.8.1}$$

$$f''(x) = \sinh(x) \tag{1.1.8.2}$$

Successive differentiation establishes that the derivative alternates with period 2. For any integer $m \geq 1$:

$$f^{(2m)}(x) = \sinh(x) \tag{1.1.8.3}$$

$$f^{(2m-1)}(x) = \cosh(x) \tag{1.1.8.4}$$

1.1.9 Relational Algebra Natural Join

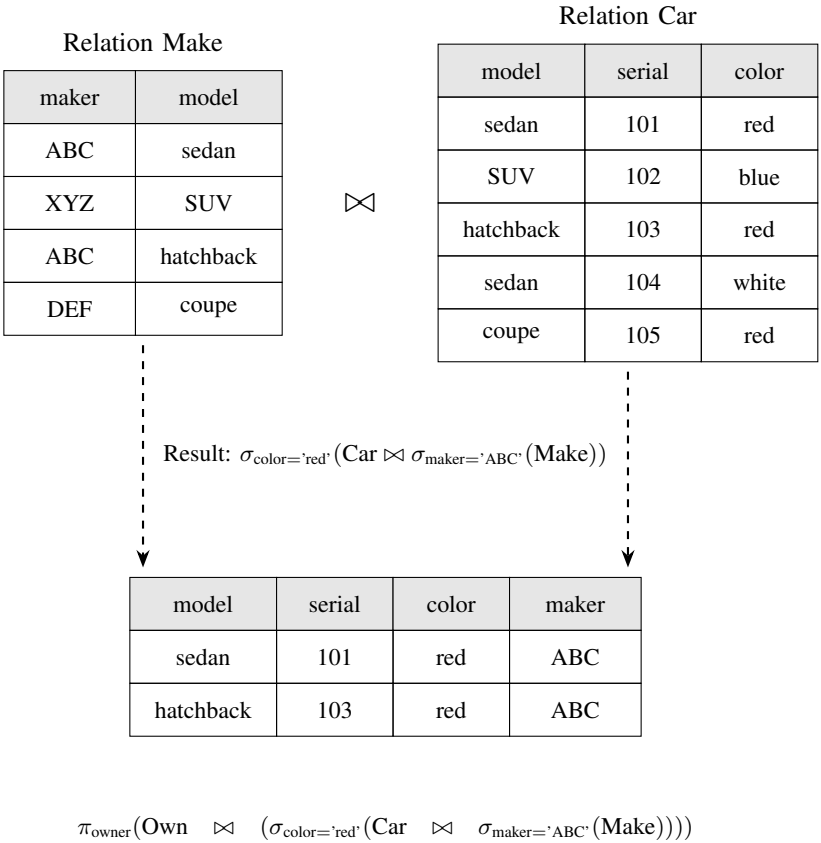


Fig. 1.1.9.1: Pictorial representation of relational joins in the Car-Make-Own database schema

$$\sigma_c(R) = \{t \in R \mid c(t)\} \tag{1.1.9.1}$$

$$R \bowtie S = \{t \cup s \mid t \in R,\ s \in S,\ t[A] = s[A]\ \forall A \in \text{attr}(R) \cap \text{attr}(S)\} \tag{1.1.9.2}$$

$$\pi_L(R) = \{t[L] \mid t \in R\} \tag{1.1.9.3}$$

Solution: The operators define the query evaluation:

- $\sigma_c(R)$ (Selection): Filters tuples in relation R that satisfy condition c (e.g., filtering by maker = 'ABC' or color = 'red').
- $R \bowtie S$ (Natural Join): Merges tuples from relations R and S having identical values on their common attributes (e.g., matching on `model` or `serial`) and eliminates duplicate attribute columns.
- $\pi_L(R)$ (Projection): Extracts only the specified attribute list L from the relation (e.g., isolating π_{owner}).

1.1.10 Consider the following three relations:

$$\text{Car}\ (\underline{\text{model}}, \underline{\text{year}}, \underline{\text{serial}}, \text{color})$$
$$\text{Make}\ (\underline{\text{maker}}, \underline{\text{model}})$$
$$\text{Own}\ (\underline{\text{owner}}, \underline{\text{serial}})$$

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; (owner, serial) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}}\left(\text{Own} \bowtie \left(\sigma_{\text{color='red'}}\left(\text{Car} \bowtie \left(\sigma_{\text{maker='ABC'}}\text{Make}\right)\right)\right)\right) \tag{1.1.10.1}$$

Which one of the following options describes what the above expression computes? (DA 2025)

i) All owners of a red car, a car made by ABC, or a red car made by ABC
ii) All owners of more than one car, where at least one car is red and made by ABC
iii) All owners of a red car made by ABC
iv) All red cars made by ABC

Solution: From (1.1.9.2), $\sigma_{\text{color='red'}}\left(\text{Car} \bowtie \left(\sigma_{\text{maker='ABC'}}\text{Make}\right)\right)$ filters red cars made by ABC. Joining with Own and projecting via π_{owner} selects:

$$\pi_{\text{owner}}\left(\text{Own} \bowtie (\dots)\right) \tag{1.1.10.2}$$

which corresponds to all owners of a red car made by ABC.
The following code verifies the solution

codes/qF10/qF10.c

1.1.11 Linear Probing in Hash Tables

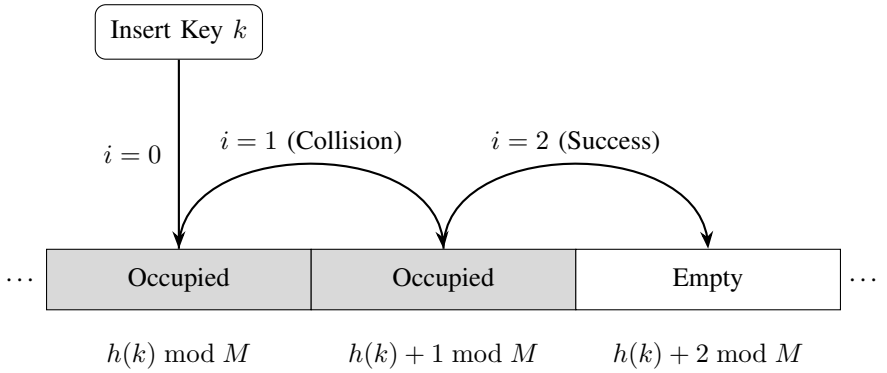


Fig. 1.1.11.1: Pictorial representation of Linear Probing in Hash Tables

$$h\left(k,i\right) = \left(h\left(k\right) + i\right) \pmod{M} \tag{1.1.11.1}$$

The terms in the probing function represent:

- k: The original key or piece of data to be stored.
- $h\left(k\right)$: The primary hash function that calculates the initial target index.
- i: The probe attempt number, starting at 0 and incrementing linearly for each collision encountered.
- M: The total allocated size of the hash table.
- $\pmod{M}$: The modulo operation ensuring the probe sequence wraps around to index 0 if the end of the table is reached.
- $h\left(k,i\right)$: The final calculated storage index for the i-th probe attempt.

1.1.12 Consider a hash table of size 10 with indices $\{0, 1, \ldots, 9\}$, with the hash function

$$h\left(x\right) = 3x \left(\text{mod } 10\right) \tag{1.1.12.1}$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: $1, 4, 5, 6, 14, 15$. The indices where the keys 14 and 15 are stored are, respectively (DA 2025)

i) 2 and 5 ii) 2 and 6 iii) 4 and 5 iv) 4 and 6

Solution: From (1.1.11.1), $h(k,i) = (3k+i) \pmod{10}$:

$$k = 1 \implies h(1,0) = 3 \tag{1.1.12.2}$$
$$k = 4 \implies h(4,0) = 2 \tag{1.1.12.3}$$
$$k = 5 \implies h(5,0) = 5 \tag{1.1.12.4}$$
$$k = 6 \implies h(6,0) = 8 \tag{1.1.12.5}$$
$$k = 14 \implies h(14,0) = 2 \tag{1.1.12.6}$$
$$\implies h(14,1) = 3 \tag{1.1.12.7}$$
$$\implies h(14,2) = 4 \tag{1.1.12.8}$$
$$k = 15 \implies h(15,0) = 5 \tag{1.1.12.9}$$
$$\implies h(15,1) = 6 \tag{1.1.12.10}$$

1.1.13 Median of a Continuous Distribution

Let $f_X(x)$ be the probability density function. The median m divides the total probability mass strictly in half:

$$\Pr(X \leq m) = 0.5 \tag{1.1.13.1}$$
$$\int_{-\infty}^{m} f_X(x)\, dx = 0.5 \tag{1.1.13.2}$$

Because the integral of $f_X(x)$ from $-\infty$ to m is exactly $F_X(m)$, it follows that:

$$F_X(m) = \int_{-\infty}^{m} f_X(x)\, dx = 0.5 \tag{1.1.13.3}$$

1.1.14 General Moments of the Standard Normal Distribution

The expected value $E[Z^n]$ for a standard normal variable $Z \sim \mathcal{N}(0,1)$ is defined by the integral:

$$E[Z^n] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^n e^{-z^2/2}\, dz \tag{1.1.14.1}$$

For odd powers ($n = 2k+1$), the integrand in (1.1.14.1) is an odd function symmetric about the origin:

$$E\left[Z^{2k+1}\right] = 0 \tag{1.1.14.2}$$

For even powers ($n = 2k$), integrating (1.1.14.1) by parts ($u = z^{n-1}$, $dv = ze^{-z^2/2}dz$) yields:

$$\int_{-\infty}^{\infty} z^n e^{-z^2/2}\, dz = \left[-z^{n-1}e^{-z^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} (n-1)\, z^{n-2} e^{-z^2/2}\, dz \tag{1.1.14.3}$$

The boundary term in (1.1.14.3) decays to zero. Multiplying by the normalization constant from (1.1.14.1) establishes the recurrence relation:

$$E[Z^n] = (n-1)\, E\left[Z^{n-2}\right] \tag{1.1.14.4}$$

Expanding (1.1.14.4) recursively to the base case $E\left[Z^0\right] = 1$ gives the expanded

product:

$$E\left[Z^n\right] = (n-1)\cdot(n-3)\cdots 3\cdot 1 \tag{1.1.14.5}$$

1.1.15 Let the random variable Z follow a standard normal distribution. It follows from (1.1.14.2) and (1.1.14.5) that

$$E[Z] = 0 \quad \text{(using (1.1.14.2) with } k=0) \tag{1.1.15.1}$$
$$E\left[Z^2\right] = 1 \quad \text{(using (1.1.14.5) with } n=2) \tag{1.1.15.2}$$

Show that for

$$X = aZ + b \tag{1.1.15.3}$$

It follows that

$$E[X] = aE[Z] + b \tag{1.1.15.4}$$
$$\text{Var}(X) = a^2\text{Var}(Z) \tag{1.1.15.5}$$

Solution: The step-by-step derivations for the expectations are:

$$E[X] = E[aZ+b] \quad \text{(using (1.1.15.3))} \tag{1.1.15.6}$$
$$= aE[Z] + b \quad \text{(using (1.1.15.4))} \tag{1.1.15.7}$$
$$= a(0) + b \quad \text{(using (1.1.15.1))} \tag{1.1.15.8}$$
$$= b \tag{1.1.15.9}$$

Then,

$$X - E[X] = (aZ+b) - b \quad \text{(using (1.1.15.3) and (1.1.15.9))} \tag{1.1.15.10}$$
$$= aZ \tag{1.1.15.11}$$

Similarly,

$$E[(X - E[X])Z] = E[aZ\cdot Z] \quad \text{(using (1.1.15.11))} \tag{1.1.15.12}$$
$$= aE\left[Z^2\right] \tag{1.1.15.13}$$
$$= a(1) \quad \text{(using (1.1.15.2))} \tag{1.1.15.14}$$
$$= a \tag{1.1.15.15}$$

Yielding,

$$E\left[(X - E[X])^2\right] = E\left[(aZ)^2\right] \quad \text{(using (1.1.15.11))} \tag{1.1.15.16}$$
$$= a^2E\left[Z^2\right] \tag{1.1.15.17}$$
$$= a^2(1) \quad \text{(using (1.1.15.2))} \tag{1.1.15.18}$$
$$= a^2 \tag{1.1.15.19}$$

1.1.16 Tail Probability of Exponential Distribution

$$\Pr(X \geq x) = \int_x^\infty \lambda e^{-\lambda t}\, dt = e^{-\lambda x} \tag{1.1.16.1}$$

1.1.17 Let $A = [a_1, a_2, \ldots, a_n]$ and $B = [b_1, b_2, \ldots, b_m]$ represent sequences, and let $\leftarrow$ denote in-place modification. The python functions map to the following mathematical transformations:

$$A.\text{extend}(B) \implies A \leftarrow [a_1, \ldots, a_n, b_1, \ldots, b_m] \tag{1.1.17.1}$$
$$A.\text{append}(B) \implies A \leftarrow [a_1, \ldots, a_n, B] \tag{1.1.17.2}$$
$$A.\text{update}(B) \implies A \leftarrow A \cup B \quad \text{(where } A \text{ and } B \text{ are sets)} \tag{1.1.17.3}$$
$$A.\text{insert}(i, B) \implies A \leftarrow [a_1, \ldots, a_i, B, a_{i+1}, \ldots, a_n] \tag{1.1.17.4}$$

1.1.18 Differentiability Rules

$$(fg)'(c) = f'(c)\, g(c) + f(c)\, g'(c) \tag{1.1.18.1}$$
$$(f \pm g)'(c) = f'(c) \pm g'(c) \tag{1.1.18.2}$$
$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)\, g(c) - f(c)\, g'(c)}{[g(c)]^2} \tag{1.1.18.3}$$

1.1.19 Properties of Orthonormal Sets

$$\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{1.1.19.1}$$

An orthonormal set of vectors in $\mathbb{R}^n$ is unconditionally linearly independent.(Note that linear independence does not imply orthogonality) Orthonormal bases in $\mathbb{R}^n$ are not unique because applying any rotation matrix generates a new valid orthonormal basis.

1.1.20 Binary Search Complexity

$$T(n) = T(n/2) + O(1) \tag{1.1.20.1}$$
$$T(n) = O(\log n) \tag{1.1.20.2}$$

Solution: Binary search operates by halving the search space at each step. Let c represent the constant $O(1)$ time required to access the middle element. Expanding the recurrence relation iteratively yields:

$$T(n) = T\left(\frac{n}{2}\right) + c \tag{1.1.20.3}$$
$$= \left(T\left(\frac{n}{4}\right) + c\right) + c \tag{1.1.20.4}$$
$$= T\left(\frac{n}{2^2}\right) + 2c \tag{1.1.20.5}$$
$$= T\left(\frac{n}{2^k}\right) + kc \tag{1.1.20.6}$$

The recursion strictly terminates when the array size reduces to 1.

$$\frac{n}{2^k} = 1 \tag{1.1.20.7}$$

$$n = 2^k \tag{1.1.20.8}$$

$$k = \log_2 n \tag{1.1.20.9}$$

Substituting k back into the expanded equation:

$$T(n) = T(1) + c \log_2 n \tag{1.1.20.10}$$

$$= O(\log n) \tag{1.1.20.11}$$

This logarithmic performance bound relies entirely on executing the $O(1)$ constant-time midpoint calculation, restricting its use strictly to contiguous memory structures like arrays, rendering it incompatible with linked lists.

1.1.21

$$A = I_n + xx^\top \tag{1.1.21.1}$$

Given the condition $x^\top x = a$, we can determine the eigenvalues by evaluating the standard eigenvalue equation $Av = \lambda v$ across the two geometric subspaces defined by x.

For an eigenvector parallel to x (let $v = x$):

$$Ax = \left(I_n + xx^\top\right) x \tag{1.1.21.2}$$

$$= x + x\left(x^\top x\right) \tag{1.1.21.3}$$

$$= x + x\,(a) \tag{1.1.21.4}$$

$$= (1 + a)\, x \tag{1.1.21.5}$$

yielding a single eigenvalue of $\lambda_1 = 1 + a$.

For any eigenvector v orthogonal to x (meaning $x^\top v = 0$):

$$Av = \left(I_n + xx^\top\right) v \tag{1.1.21.6}$$

$$= v + x\left(x^\top v\right) \tag{1.1.21.7}$$

$$= v + x\,(0) \tag{1.1.21.8}$$

$$= 1v \tag{1.1.21.9}$$

yielding an eigenvalue of $\lambda_2 = 1$ with multiplicity $n - 1$.

To derive the corresponding eigenvalues for the inverse matrix A^{-1}, we use the defining property of eigenvalues $Av = \lambda v$ and left-multiply both sides by A^{-1}:

$$A^{-1}(Av) = A^{-1}(\lambda v) \tag{1.1.21.10}$$

$$\left(A^{-1}A\right) v = \lambda\left(A^{-1}v\right) \tag{1.1.21.11}$$

$$I_n v = \lambda A^{-1} v \tag{1.1.21.12}$$

$$v = \lambda A^{-1} v \tag{1.1.21.13}$$

$$A^{-1} v = \frac{1}{\lambda} v \tag{1.1.21.14}$$

This proves that any eigenvector v of A is also an eigenvector of A^{-1}, and its corresponding eigenvalue is exactly the reciprocal $\frac{1}{\lambda}$. Substituting our previously derived values, the eigenvalues of A^{-1} are $\frac{1}{1+a}$ and 1.

1.1.22 Insertion Sort Inversions

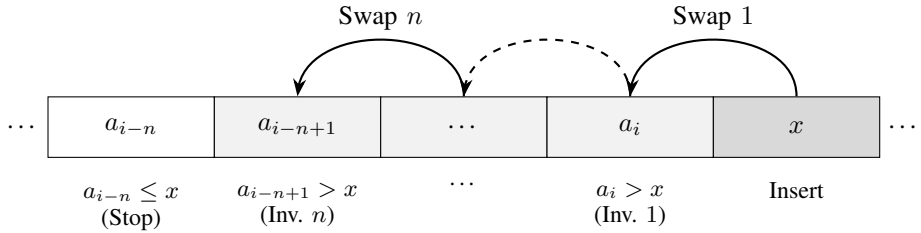


Fig. 1.1.22.1: Pictorial representation of Insertion Sort executing exactly n swaps for n inversions

$$\text{Swaps} = \text{Number of Inversions} \tag{1.1.22.1}$$

Solution: Insertion sort executes exactly one swap per inverted pair. Therefore, for the array to achieve a completely sorted state using exactly n swaps, the newly inserted element x must be strictly smaller than exactly n preceding elements.

1.1.23 Bayes' Theorem:

$$Pr\left(E_2|W\right) = \frac{Pr\left(E_2\right) Pr\left(W|E_2\right)}{\sum_i Pr\left(E_i\right) Pr\left(W|E_i\right)} \tag{1.1.23.1}$$

1.1.24 Asymptotic Radical Limits

$$\lim_{t\to\infty} \left(\sqrt{t^2+at} - t\right) = \frac{a}{2} \tag{1.1.24.1}$$

Solution: Multiplying by the conjugate:

$$\lim_{t\to+\infty} \frac{\left(t^2+at\right) - t^2}{\sqrt{t^2+at}+t} = \lim_{t\to+\infty} \frac{a}{\sqrt{1+\frac{a}{t}}+1} = \frac{a}{2} \tag{1.1.24.2}$$

1.1.25 Setting up MySQL (MariaDB) and Executing '.sql' Files on Termux

To install, initialize, run the MariaDB daemon, and execute external '.sql' script files inside the Termux environment on Android:

- Initialize Packages & Install MariaDB:

```
# Grant storage access to Termux
termux-setup-storage

# Update mirrors and install mariadb engine
pkg update && pkg upgrade -y
pkg install mariadb -y
```

- Initialize Data Directory & Start Server Daemon:

```
# Initialize MySQL system tables
mariadb-install-db

# Start MariaDB server daemon in the background
mysqld_safe &
```

- Interactive Console Execution (‘SOURCE‘ Command):

```
-- Connect to MariaDB console
mariadb -u root

-- Execute .sql script inside interactive shell
SOURCE /sdcard/Download/codes/q40/script.sql;

-- Alternative shortcut notation:
\. /sdcard/Download/codes/q40/script.sql
```

1.1.26 On a relation named Loan of a bank:

Loan		
loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Fig. 1.1.26.1

the following SQL query is executed.

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
        FROM Loan L2
        WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is _____. (DA 2025)

Solution: The inner subquery evaluates the maximum loan amount for the ’SR

`Nagar'` branch, giving $\max\{40000, 25000, 65000\} = 65000$. The outer query selects candidate rows satisfying amount > 65000, which matches $\{\text{L11}, \text{L23}, \text{L25}\}$.

$$\text{Number of rows returned} = 3 \tag{1.1.26.1}$$

The following code verifies the solution

codes/qF25/qF25.sql

1.1.27 Least Squares Regression Through Origin (Matrix Form)

$$w^* = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y} \tag{1.1.27.1}$$

Solution: For a linear model constrained to pass strictly through the origin, $\hat{y}_i = wx_i$. Given n observations, the design vector $\mathbf{X}$ and target vector $\mathbf{y}$ are expressed as:

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \tag{1.1.27.2}$$

The Sum of Squared Residuals in vector-matrix notation is given by $S(w) = \|\mathbf{y} - \mathbf{X}w\|^2$. Expanding and differentiating with respect to w, we set the derivative to zero to derive the normal equations:

$$\frac{d}{dw}\left(\mathbf{y}^\top \mathbf{y} - 2w\mathbf{X}^\top \mathbf{y} + w^2 \mathbf{X}^\top \mathbf{X}\right) = 0 \tag{1.1.27.3}$$

$$-2\mathbf{X}^\top \mathbf{y} + 2w\mathbf{X}^\top \mathbf{X} = 0 \tag{1.1.27.4}$$

$$\mathbf{X}^\top \mathbf{X}\, w = \mathbf{X}^\top \mathbf{y} \tag{1.1.27.5}$$

$$w^* = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y} \tag{1.1.27.6}$$

1.1.28 Bayes Optimal Error Rate

$$Pr\left(\text{error} \mid \mathbf{x}\right) = \frac{\min_k Pr\left(\mathbf{x} \mid y_k\right) Pr\left(y_k\right)}{Pr\left(\mathbf{x}\right)} \tag{1.1.28.1}$$

1.1.29 Moments of Standard Normal

$$Z \sim \mathcal{N}(0,1) \implies E\left[Z^2\right] = 1, E\left[Z^4\right] = 3 \quad \text{(using (1.1.14.5))} \tag{1.1.29.1}$$

$$var\left(Y\right) = E\left[Y^2\right] - \left(E\left[Y\right]\right)^2 \tag{1.1.29.2}$$

1.1.30 Rank of Involutory Matrices

$$\mathbf{A}^3 = \mathbf{A} \implies \text{rank}\left(\mathbf{A}\right) = \text{rank}\left(\mathbf{A}^2\right) \tag{1.1.30.1}$$

Solution: Let $\mathbf{z} \in \text{Col}\left(\mathbf{A}^2\right)$. By definition, there exists a vector $\mathbf{v}$ such that:

$$\mathbf{z} = \mathbf{A}^2 \mathbf{v} \tag{1.1.30.2}$$

$$= \mathbf{A}\left(\mathbf{A}\mathbf{v}\right) \tag{1.1.30.3}$$

Letting $\mathbf{u} = \mathbf{A}\mathbf{v}$, we have $\mathbf{z} = \mathbf{A}\mathbf{u}$, which dictates $\mathbf{z} \in \text{Col}\,(\mathbf{A})$. Thus:

$$\text{Col}\left(\mathbf{A}^2\right) \subseteq \text{Col}\,(\mathbf{A}) \implies \text{rank}\left(\mathbf{A}^2\right) \leq \text{rank}\,(\mathbf{A}) \tag{1.1.30.4}$$

Conversely, let $\mathbf{y} \in \text{Col}\,(\mathbf{A})$. Then there exists a vector $\mathbf{x}$ such that:

$$\mathbf{y} = \mathbf{A}\mathbf{x} \tag{1.1.30.5}$$
$$= \mathbf{A}^3\mathbf{x} \quad (\text{given } \mathbf{A} = \mathbf{A}^3) \tag{1.1.30.6}$$
$$= \mathbf{A}^2\,(\mathbf{A}\mathbf{x}) \tag{1.1.30.7}$$

This demonstrates $\mathbf{y} \in \text{Col}\left(\mathbf{A}^2\right)$, establishing the reverse inclusion:

$$\text{Col}\,(\mathbf{A}) \subseteq \text{Col}\left(\mathbf{A}^2\right) \implies \text{rank}\,(\mathbf{A}) \leq \text{rank}\left(\mathbf{A}^2\right) \tag{1.1.30.8}$$

Consequently, the ranks must be equal:

$$\text{rank}\,(\mathbf{A}) = \text{rank}\left(\mathbf{A}^2\right) \tag{1.1.30.9}$$

1.1.31 Construct the matrix $\mathbf{X} \in \mathbb{R}^{n\times n}$ having the given vectors as its columns:

$$\mathbf{X} = \begin{pmatrix}\mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_n\end{pmatrix} \tag{1.1.31.1}$$

By definition of matrix multiplication, the (i,j)-th entry of $\mathbf{X}^\top\mathbf{X}$ is $\mathbf{x}_i^\top\mathbf{x}_j$. Therefore:

$$\mathbf{A} = \mathbf{X}^\top\mathbf{X} \tag{1.1.31.2}$$

Notice that $\mathbf{A}$ is symmetric since $\mathbf{A}^\top = \left(\mathbf{X}^\top\mathbf{X}\right)^\top = \mathbf{X}^\top\mathbf{X} = \mathbf{A}$.
For any arbitrary non-zero vector $\mathbf{z} \in \mathbb{R}^n$, the quadratic form evaluates to:

$$\mathbf{z}^\top\mathbf{A}\mathbf{z} = \mathbf{z}^\top\left(\mathbf{X}^\top\mathbf{X}\right)\mathbf{z} \tag{1.1.31.3}$$
$$= (\mathbf{X}\mathbf{z})^\top(\mathbf{X}\mathbf{z}) \tag{1.1.31.4}$$
$$= \|\mathbf{X}\mathbf{z}\|^2 \tag{1.1.31.5}$$

Since the columns $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ are linearly independent, the linear combination satisfies:

$$\mathbf{X}\mathbf{z} = \sum_{k=1}^{n} z_k\mathbf{x}_k = \mathbf{0} \iff \mathbf{z} = \mathbf{0} \tag{1.1.31.6}$$

1.1.32 Interval Probabilities from CDF

$$\Pr\left(X^2 \leq c^2\right) = F_X\,(c) - F_X\,(-c) \tag{1.1.32.1}$$

Solution: For any constant $c > 0$, the quadratic inequality $X^2 \leq c^2$ algebraically resolves to a bounded symmetric interval. The probability of this event is evaluated by separating the interval bounds:

$$\Pr\left(X^2 \leq c^2\right) = \Pr\,(-c \leq X \leq c) \tag{1.1.32.2}$$
$$= \Pr\,(X > -c \text{ and } X \leq c) \tag{1.1.32.3}$$
$$= \Pr\,(X \leq c) - \Pr\,(X \leq -c) \tag{1.1.32.4}$$

Substituting the fundamental definition of the Cumulative Distribution Function, $F_X(x) = \Pr(X \leq x)$ (assuming a continuous distribution where $\Pr(X = -c) = 0$), yields the final expression:

$$\Pr\left(X^2 \leq c^2\right) = F_X(c) - F_X(-c) \tag{1.1.32.5}$$

1.1.33 Central Limit Theorem Approximation for Binomial Sums

For independent and identically distributed random variables $X_i \sim \text{Bernoulli}(\theta)$, the sum $Y \sim \text{Bin}(n, \theta)$, its mean μ, and standard deviation σ are given by:

$$Y = \sum_{i=1}^{n} X_i \tag{1.1.33.1}$$

$$\mu = n\theta \tag{1.1.33.2}$$

$$\sigma = \sqrt{n\theta(1-\theta)} \tag{1.1.33.3}$$

For large n such that $n\theta \geq 5$ and $n(1-\theta) \geq 5$, the binomial sum Y is approximated by a Gaussian distribution $\mathcal{N}\left(\mu, \sigma^2\right)$ via the Central Limit Theorem:

$$Z = \frac{Y - \mu}{\sigma} = \frac{Y - n\theta}{\sqrt{n\theta(1-\theta)}} \xrightarrow{d} \mathcal{N}(0,1) \tag{1.1.33.4}$$

Standardizing the interval $[a, b]$ maps the boundaries to the standard normal variable Z. Without the 0.5 continuity correction factor applied, the interval probability evaluates directly in terms of the standard normal Cumulative Distribution Function (CDF), denoted by $\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$:

$$\begin{aligned} P(a \leq Y \leq b) &\approx P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right) \\ &= \phi\left(\frac{b - n\theta}{\sqrt{n\theta(1-\theta)}}\right) - \phi\left(\frac{a - n\theta}{\sqrt{n\theta(1-\theta)}}\right) \end{aligned} \tag{1.1.33.5}$$

1.1.34

$$\Pr(\lfloor X \rfloor = \ell) = e^{-\lambda \ell}\left(1 - e^{-\lambda}\right) \tag{1.1.34.1}$$

Solution: Let X be an exponentially distributed random variable with rate parameter $\lambda = \frac{1}{\mu}$. Its probability density function is $f_X(x) = \lambda e^{-\lambda x}$.

The event $Y = \lfloor X \rfloor = \ell$ strictly occurs when X falls within the bounded interval $[\ell, \ell + 1)$. The probability is derived by integrating the PDF over these boundaries:

$$P(Y = \ell) = P(\ell \leq X < \ell + 1) \tag{1.1.34.2}$$

$$= \int_{\ell}^{\ell+1} \lambda e^{-\lambda x}\, dx \tag{1.1.34.3}$$

$$= \left[-e^{-\lambda x}\right]_{\ell}^{\ell+1} \tag{1.1.34.4}$$

$$= e^{-\lambda \ell} - e^{-\lambda(\ell+1)} \tag{1.1.34.5}$$

$$= e^{-\lambda \ell}\left(1 - e^{-\lambda}\right) \tag{1.1.34.6}$$

By substituting $q = e^{-\lambda}$, the exact geometric probability mass function emerges:

$$P(Y = \ell) = q^{\ell}(1 - q) \tag{1.1.34.7}$$

1.1.35 Probability of At Least One Success via Z-Transform

$$M_X(z) = \left(q + pz^{-1}\right)^n \tag{1.1.35.1}$$

Let $X = \sum_{i=1}^{n} X_i$ represent the total sum of n independent trials, where each $X_i \sim$ Bernoulli (p). The Z-transform for a single trial is $M_{X_i}(z) = q + pz^{-1}$, and the aggregate transform is:

$$M_X(z) = \left(q + pz^{-1}\right)^n \tag{1.1.35.2}$$

Expanding this polynomial yields the probability mass function $p_X(k)$ as the coefficient of the z^{-k} term. The probability of zero successes ($k = 0$) corresponds to the constant term q^n. Applying the complement rule to find the probability of at least one success:

$$P(X \geq 1) = 1 - p_X(0) \tag{1.1.35.3}$$

$$= 1 - q^n \tag{1.1.35.4}$$

1.1.36

$$f'(x) = 0 \text{ at local extrema} \tag{1.1.36.1}$$

1.1.37 Properties of Orthogonal Projection Matrices

$$\mathbf{A} = \sum_{i=1}^{k} \mathbf{x}_i \mathbf{x}_i^{\top} \implies \mathbf{A}^2 = \mathbf{A} \tag{1.1.37.1}$$

Solution: For an orthonormal set of vectors where the inner product $\mathbf{x}_i^{\top}\mathbf{x}_j = \delta_{ij}$:

$$\mathbf{A}^2 = \left(\sum_{i=1}^{k} \mathbf{x}_i \mathbf{x}_i^{\top}\right)\left(\sum_{j=1}^{k} \mathbf{x}_j \mathbf{x}_j^{\top}\right) \tag{1.1.37.2}$$

$$= \sum_{i=1}^{k}\sum_{j=1}^{k} \mathbf{x}_i \left(\mathbf{x}_i^{\top}\mathbf{x}_j\right)\mathbf{x}_j^{\top} \tag{1.1.37.3}$$

$$= \sum_{i=1}^{k} \mathbf{x}_i (1)\, \mathbf{x}_i^{\top} \tag{1.1.37.4}$$

$$= \mathbf{A} \tag{1.1.37.5}$$

1.1.38 Rolle's Theorem

$$\begin{aligned} &f(a) = f(b),\ f \text{ continuous on } [a, b],\ f \text{ differentiable on } (a, b) \\ &\implies \exists\, c \in (a, b) \text{ s.t. } f'(c) = 0 \end{aligned} \tag{1.1.38.1}$$

Rolle's theorem states that if a real-valued function is continuous on a closed interval and differentiable on the open interval, and has equal values at the endpoints, then

there exists at least one interior point where the first derivative is zero.

1.1.39 Isometric Norm Preservation

$$\|\mathbf{A}\mathbf{x}\|^2 = \|\mathbf{x}\|^2 \iff \mathbf{A}^\top \mathbf{A} = \mathbf{I} \tag{1.1.39.1}$$

Solution: By the definition of the Euclidean norm in terms of the standard inner product, evaluate both sides for an arbitrary vector $\mathbf{x} \in \mathbb{R}^n$:

$$\|\mathbf{A}\mathbf{x}\|^2 = (\mathbf{A}\mathbf{x})^\top (\mathbf{A}\mathbf{x}) = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} \tag{1.1.39.2}$$

$$\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{x} = \mathbf{x}^\top \mathbf{I}\mathbf{x} \tag{1.1.39.3}$$

Equating the two expanded forms for all $\mathbf{x}$:

$$\mathbf{x}^\top \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{x}^\top \mathbf{I}\mathbf{x} \tag{1.1.39.4}$$

$$\mathbf{x}^\top \left(\mathbf{A}^\top \mathbf{A} - \mathbf{I}\right) \mathbf{x} = 0 \tag{1.1.39.5}$$

The matrix $\left(\mathbf{A}^\top \mathbf{A} - \mathbf{I}\right)$ is inherently symmetric. If a quadratic form of a symmetric matrix evaluates to zero for all possible vectors $\mathbf{x}$, the matrix must be the zero matrix. Therefore:

$$\mathbf{A}^\top \mathbf{A} - \mathbf{I} = \mathbf{0} \implies \mathbf{A}^\top \mathbf{A} = \mathbf{I} \tag{1.1.39.6}$$

1.1.40 Hard-Margin Support Vector Machine (SVM) Formulation

- Canonical Separating Hyperplane and Decision Rule: For linearly separable data with labels $y_i \in \{-1, +1\}$, the decision boundary is $\mathbf{w}^\top \mathbf{x} + b = 0$, giving the decision rule:

$$f(\mathbf{x}) = \text{sign}\left(\mathbf{w}^\top \mathbf{x} + b\right) \tag{1.1.40.1}$$

  The canonical constraints require:

$$y_i \left(\mathbf{w}^\top \mathbf{x}_i + b\right) \geq 1, \quad \forall i \in \{1, \ldots, N\} \tag{1.1.40.2}$$

- Support Vectors: A training sample $\mathbf{x}_i$ is a support vector if and only if it lies directly on its class margin boundary:

$$y_i \left(\mathbf{w}^\top \mathbf{x}_i + b\right) = 1 \tag{1.1.40.3}$$

- Geometric Margin: The total perpendicular separation between the bounding planes $\mathbf{w}^\top \mathbf{x} + b = +1$ and $\mathbf{w}^\top \mathbf{x} + b = -1$ is:

$$\gamma = \frac{2}{\|\mathbf{w}\|} \tag{1.1.40.4}$$

1.1.41 Properties of Sample Proportion

$$E[\hat{p}] = p \tag{1.1.41.1}$$

$$var(\hat{p}) = \frac{p(1-p)}{n} \tag{1.1.41.2}$$

Solution: Let $X = \sum_{i=1}^{n} X_i$ represent the total number of heads. Because X_i are independent Bernoulli random variables, their sum follows a Binomial distribution

$X \sim B(n,p)$.
From the defined moments of a Binomial distribution:

$$E(X) = np \tag{1.1.41.3}$$
$$var(X) = npq = np(1-p) \tag{1.1.41.4}$$

The sample proportion is given as $\hat{p} = \frac{X}{n}$. Evaluating the expectation yields:

$$E[\hat{p}] = E\left[\frac{X}{n}\right] \tag{1.1.41.5}$$
$$= \frac{1}{n}E(X) \tag{1.1.41.6}$$
$$= \frac{1}{n}(np) = p \tag{1.1.41.7}$$

To evaluate the variance, apply the fundamental variance scaling property $var(aX) = a^2 var(X)$ established in the text:

$$var(\hat{p}) = var\left(\frac{X}{n}\right) \tag{1.1.41.8}$$
$$= \frac{1}{n^2}var(X) \tag{1.1.41.9}$$
$$= \frac{1}{n^2}[np(1-p)] \tag{1.1.41.10}$$
$$= \frac{p(1-p)}{n} \tag{1.1.41.11}$$

Because n is in the denominator of the resulting expression, the variance of $\hat{p}$ strictly decreases as the total number of independent coin tosses n increases.

1.1.42 Shortest Path Metric

$$d(u,v) \leq d(u,w) + d(w,v) \tag{1.1.42.1}$$

Fake edges violating the triangle inequality are identifiable by calculating the minimum spanning distances directly.

1.1.43 Lipschitz Continuity with Exponent > 1

$$|f(x) - f(y)| \leq K|x-y|^{\alpha} \implies f'(x) = 0 \tag{1.1.43.1}$$

Solution: For $y \neq x$, dividing both sides by $|x-y|$:

$$\left|\frac{f(x)-f(y)}{x-y}\right| \leq K|x-y|^{\alpha-1} \tag{1.1.43.2}$$

Taking the limit as $y \to x$ for $\alpha > 1$ (where $\alpha - 1 > 0$):

$$|f'(x)| = \lim_{y\to x}\left|\frac{f(x)-f(y)}{x-y}\right| \tag{1.1.43.3}$$
$$\leq \lim_{y\to x} K|x-y|^{\alpha-1} = 0 \tag{1.1.43.4}$$
$$\implies f'(x) = 0 \tag{1.1.43.5}$$

Since $f'(x) = 0$ for all x, $f(x)$ is a constant function.

1.1.44 Projected Variance and Principal Components

$$\mathbf{C} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}^{(i)}\mathbf{x}^{(i)\top} \quad \text{(Covariance Matrix for centered data)} \tag{1.1.44.1}$$

$$\sigma_{\mathbf{u}}^2 = \frac{1}{n}\sum_{i=1}^{n} \left(\mathbf{u}^\top \mathbf{x}^{(i)}\right)^2 = \mathbf{u}^\top \mathbf{C}\mathbf{u} \quad \text{(Variance along direction } \mathbf{u}\text{)} \tag{1.1.44.2}$$

$$\max_{\mathbf{u}^\top\mathbf{u}=1} \mathbf{u}^\top \mathbf{C}\mathbf{u} = \lambda_{\max} \quad \text{(Rayleigh Quotient / Maximum Variance)} \tag{1.1.44.3}$$

1.1.45 Expectation of Indicator Sums

$$E\left[\sum_{i=1}^{n} I_i\right] = \sum_{i=1}^{n} E\left[I_i\right] = np \tag{1.1.45.1}$$

Linearity of expectation holds uniformly irrespective of variable dependence or independence.

1.1.46 Relational Division

R **(Student, Skill)**

A	B
Alice	SQL
Alice	Java
Alice	Python
Bob	SQL
Bob	Python
Charlie	SQL
Charlie	Java
Charlie	Python

$\div$

S **(Required)**

B
SQL
Java
Python

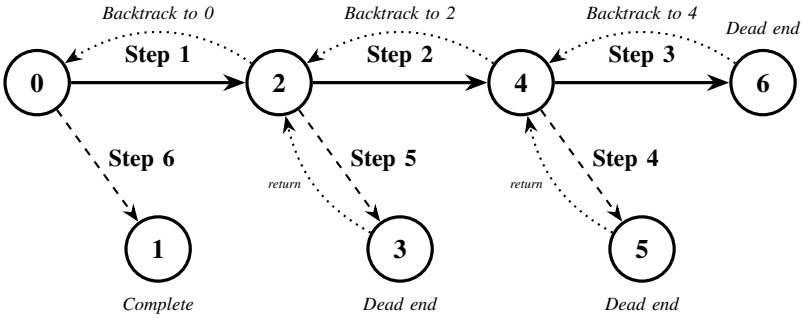
$=$

$R \div S$

A
Alice
Charlie

$$R \div S = \pi_A(R) - \pi_A\left((\pi_A(R) \times S) - R\right) \tag{1.1.46.1}$$

1.1.47 Depth-First Search (DFS) Traversal Sequence



$$V = \{0, 1, \ldots, n\}, \quad E = \{(i, j) : 0 < j - i \leq 2\} \tag{1.1.47.1}$$

Solution: Decreasing adjacency order restricts the neighbor queue to $Adj(i) = [i + 2, i + 1]$. A DFS from 0 greedily traverses even vertices forward via $i + 2$ edges, then odd vertices during backtracking via $i + 1$ edges. For an even boundary n, the vertex $D(k)$ discovered at step $k \in \{0, \ldots, n\}$ is:

$$D(k) = \begin{cases} 2k & 0 \leq k \leq n/2 \quad \text{(Forward)} \\ 2n - 2k + 1 & n/2 < k \leq n \quad \text{(Backtrack)} \end{cases} \tag{1.1.47.2}$$

To find the number of vertices discovered chronologically after a target v, solve $D(k_v) = v$ for its discovery index k_v. Out of $n + 1$ total vertices, the remaining unvisited count evaluates to:

$$\text{Remaining Discoveries} = n - k_v \tag{1.1.47.3}$$

B. Problems

1.2.1 A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is: (DA 2025)

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

Fig. 1.2.1.1: Pixel Grid

i) 3 ii) 8 iii) 11 iv) 9

Solution: From (1.1.1.1) and (1.1.1.2), for threshold $c = 4$:

$$N = \sum_{i=1}^{4} \sum_{j=1}^{4} \mathbb{I}\left(U_{ij}\right) \tag{1.2.1.1}$$

$$= 11 \tag{1.2.1.2}$$

The following code verifies the solution

codes/q1/q1.c

1.2.2 A rectangle has a length L and a width W, where $L > W$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W? (DA 2025)

i) Perimeter increases by 10%.
ii) Length of the diagonals increases by 10%.
iii) Area increases by 10%.
iv) The rectangle becomes a square.

Solution: From (1.1.2.1), with width scaled by $\alpha = 0.10$ ($W' = 1.10W$):

$$A' = 1.10A \tag{1.2.2.1}$$

$$P' = 2\left(L + 1.10W\right) \neq 1.10 \times 2\left(L + W\right) \tag{1.2.2.2}$$

$$D' = \sqrt{L^2 + \left(1.10W\right)^2} \neq 1.10\sqrt{L^2 + W^2} \tag{1.2.2.3}$$

Furthermore, for arbitrary $L > W$ (e.g., $L = 10, W = 1$), $W' = 1.10 \neq L$, so the rectangle does not generally become a square. Thus, only the area increases by 10% for all values of L and W.

The following code verifies the solution

codes/q2/q2.c

1.2.3 Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight? (DA 2025)

i) $a_1 < a_2 < 50$
ii) $a_1 < 50 < a_2$
iii) $50 < a_1 < a_2$
iv) Either $a_1 = 50$ or $a_2 = 50$

Solution: From (1.1.3.1), for peak $c = 50$:

$$f\left(a_1\right) = f\left(a_2\right) \tag{1.2.3.1}$$

$$\implies a_1 < 50 < a_2 \tag{1.2.3.2}$$

1.2.4 If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is: (DA 2025)

i) 2^{-1} ii) 2^0 iii) 2^3 iv) 2^{15}

Solution: From (1.1.4.1):

$$3^{x^2} = 3^3 \cdot \left(3^2\right)^x \tag{1.2.4.1}$$
$$= 3^{2x+3} \tag{1.2.4.2}$$
$$\implies x^2 - 2x = 3 \tag{1.2.4.3}$$

Applying (1.1.4.1):

$$\frac{2^{x^2}}{\left(2^x\right)^2} = \frac{2^{x^2}}{2^{2x}} \tag{1.2.4.4}$$
$$= 2^{x^2-2x} \tag{1.2.4.5}$$
$$= 2^3 \tag{1.2.4.6}$$

The following code verifies the solution

codes/q4/q4.c

1.2.5 Suppose X and Y are random variables. The conditional expectation of X given Y is denoted by $E[X \mid Y]$. Then $E[E[X \mid Y]]$ equals (DA 2025)

i) $E[X \mid Y]$ ii) $\frac{E[X]}{E[Y]}$ iii) $E[X]$ iv) $E[Y]$

Solution: From (1.1.5.4) and (1.1.5.3):

$$E[E[X \mid Y]] = \sum_y E[X \mid Y = y] p_Y(y) \tag{1.2.5.1}$$
$$= \sum_y \sum_x x p_{X|Y}(x \mid y) p_Y(y) \tag{1.2.5.2}$$
$$= \sum_y \sum_x x \left(\frac{p_{X,Y}(x,y)}{p_Y(y)}\right) p_Y(y) \tag{1.2.5.3}$$
$$= \sum_y \sum_x x p_{X,Y}(x,y) \tag{1.2.5.4}$$
$$= \sum_x x \sum_y p_{X,Y}(x,y) \tag{1.2.5.5}$$
$$= \sum_x x p_X(x) \tag{1.2.5.6}$$
$$= E[X] \tag{1.2.5.7}$$

The following code verifies the solution

codes/q5/q5.c

1.2.6 The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order (DA 2025)

i) $O(n)$ ii) $O\left(n^2\right)$ iii) $O\left(n^3\right)$ iv) $O\left(n^4\right)$

Solution: From (1.1.6.1):

$$x_i = \frac{1}{U_{ii}} \left(b_i - \sum_{j=i+1}^{n} U_{ij} x_j \right) \tag{1.2.6.1}$$

For each row i ($1 \leq i \leq n$), the back-substitution step requires:

- Multiplications: $n - i$
- Additions/Subtractions: $n - i$
- Divisions: 1

Summing these operations across all n rows:

$$\text{Multiplications} = \sum_{i=1}^{n} (n - i) = \frac{n\,(n-1)}{2} \tag{1.2.6.2}$$

$$\text{Additions} = \sum_{i=1}^{n} (n - i) = \frac{n\,(n-1)}{2} \tag{1.2.6.3}$$

$$\text{Divisions} = \sum_{i=1}^{n} 1 = n \tag{1.2.6.4}$$

The total sum of all arithmetic operations evaluates to:

$$\text{Total Operations} = \frac{n\,(n-1)}{2} + \frac{n\,(n-1)}{2} + n = n^2 \tag{1.2.6.5}$$

$$= O\left(n^2\right) \tag{1.2.6.6}$$

The following code and data verifies the solution:

codes/q6/q6.c

codes/q6/timing_results.dat

The following code generates Fig. 1.2.6.1.

codes/q6/q6.py

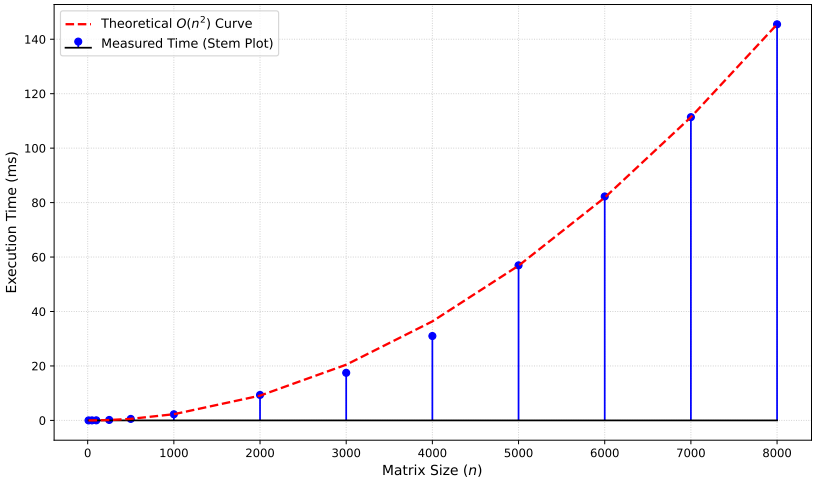


Fig. 1.2.6.1: Back-Substitution Time Complexity

1.2.7 The sum of the elements in each row of $\mathbf{A} \in \mathbb{R}^{n\times n}$ is 1. If $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$, which one of the following statements is correct (for $\mathbf{x} \in \mathbb{R}^n$)? (DA 2025)

i) The equation $\mathbf{Bx} = \mathbf{0}$ has no solution
ii) The equation $\mathbf{Bx} = \mathbf{0}$ has exactly two solutions
iii) The equation $\mathbf{Bx} = \mathbf{0}$ has infinitely many solutions
iv) The equation $\mathbf{Bx} = \mathbf{0}$ has a unique solution

Solution: From (1.1.7.1), $\mathbf{A1} = \mathbf{1}$. For $\mathbf{B} = \mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}$:

$$\begin{aligned}\mathbf{B1} &= \left(\mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A}\right)\mathbf{1} && (1.2.7.1)\\ &= \mathbf{A}^3\mathbf{1} - 2\mathbf{A}^2\mathbf{1} + \mathbf{A1} && (1.2.7.2)\\ &= \mathbf{1} - 2\mathbf{1} + \mathbf{1} && (1.2.7.3)\\ &= \mathbf{0} && (1.2.7.4)\end{aligned}$$

Thus, $\lambda = 0$ is an eigenvalue of $\mathbf{B}$, implying $\mathbf{B}$ has dependent column vectors, which yields infinitely many solutions.
The following code verifies the solution

codes/q7/q7.py

1.2.8 Let $f(x) = \frac{e^x - e^{-x}}{2}$, $x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the kth derivative of f evaluated at a. What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial) (DA 2025)

i) 0 ii) 1 iii) $\frac{1}{10!}$ iv) $\frac{2}{10!}$

Solution: From (1.1.8.3) with $m = 5$:

$$f^{(10)}(x) = \sinh(x) \tag{1.2.8.1}$$

$$\implies f^{(10)}(0) = \sinh(0) = 0 \tag{1.2.8.2}$$

1.2.9 Let X be a continuous random variable whose cumulative distribution function (CDF) $F_X(x)$, for some t, is given as follows:

$$F_X(x) = \begin{cases} 0 & x \le t \\ \frac{x-t}{4-t} & t \le x \le 4 \\ 1 & x \ge 4 \end{cases} \tag{1.2.9.1}$$

If the median of X is 3, then what is the value of t? (DA 2025)

i) 2 ii) 1 iii) -1 iv) 0

Solution: From (1.1.13.3), the median $m = 3$ satisfies:

$$F_X(3) = 0.5 \tag{1.2.9.2}$$

$$\frac{3-t}{4-t} = 0.5 \tag{1.2.9.3}$$

$$3 - t = 2 - 0.5t \tag{1.2.9.4}$$

$$\implies t = 2 \tag{1.2.9.5}$$

The following code helps generate data to plot the simulated CDF for a large sampling size

codes/q9/q9.c

The following code generates Fig. 1.2.9.1.

codes/q9/q9.py

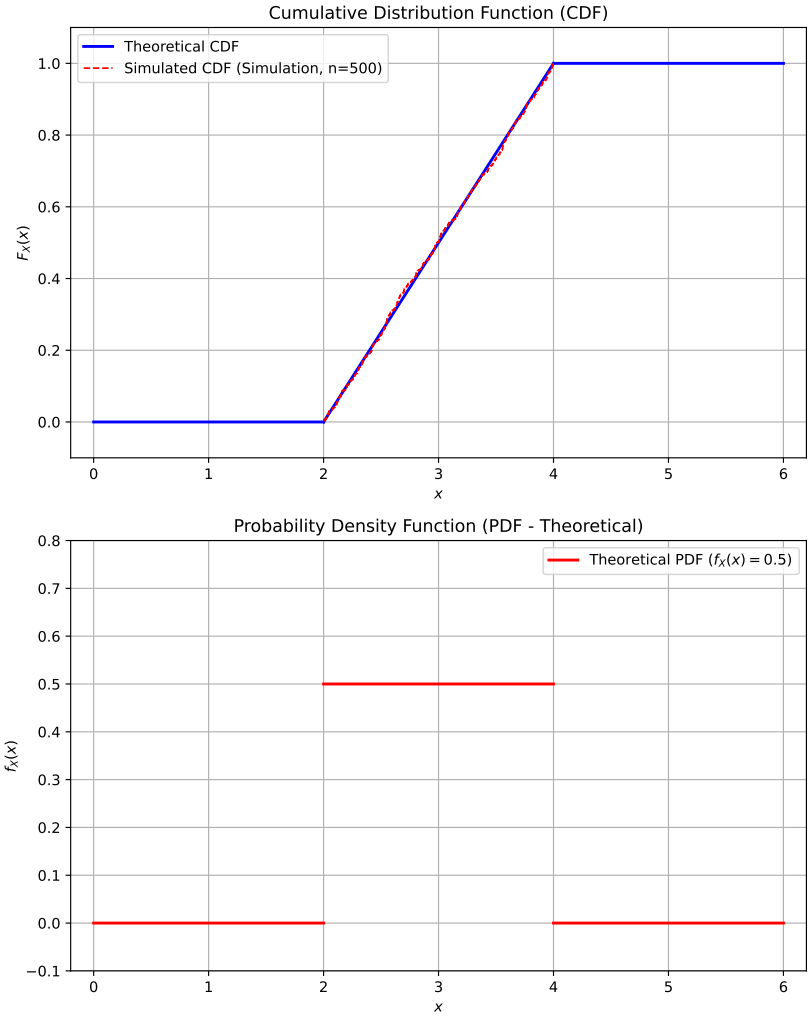


Fig. 1.2.9.1: PDF and CDF of continuous random variable X with $t = 2$

1.2.10 Let $X = aZ + b$, where Z is a standard normal random variable, and a, b are two unknown constants. It is given that

$$E[X] = 1, \quad E[(X - E[X])Z] = -2, \quad E\left[(X - E[X])^2\right] = 4, \qquad (1.2.10.1)$$

where $E[X]$ denotes the expectation of random variable X. The values of a, b are: (DA 2025)

i) $a = -2,\ b = 1$
ii) $a = 2,\ b = -1$
iii) $a = -2,\ b = -1$
iv) $a = 1,\ b = 1$

Solution: From (1.1.15.1) and (1.1.15.5), for $X = aZ + b$:

$$E[X] = aE[Z] + b \tag{1.2.10.2}$$
$$\implies b = 1 \tag{1.2.10.3}$$

From (1.1.15.2):

$$E[(X - E[X])Z] = E\left[aZ^2\right] \tag{1.2.10.4}$$
$$= a(1) \tag{1.2.10.5}$$
$$\implies a = -2 \tag{1.2.10.6}$$

The following code verifies the solution

codes/q10/q10.c

1.2.11 It is given that $P(X \geq 2) = 0.25$ for an exponentially distributed random variable X with $E[X] = \frac{1}{\lambda}$, where $E[X]$ denotes the expectation of X. What is the value of λ? ($\ln$ denotes natural logarithm) (DA 2025)

i) $\ln 2$ ii) $\ln 4$ iii) $\ln 3$ iv) $\ln 0.25$

Solution: From (1.1.16.1):

$$\Pr(X \geq 2) = e^{-2\lambda} = 0.25 = \frac{1}{4} \tag{1.2.11.1}$$
$$\implies -2\lambda = \ln\left(\frac{1}{4}\right) = -\ln 4 \tag{1.2.11.2}$$
$$\implies 2\lambda = 2\ln 2 \tag{1.2.11.3}$$
$$\implies \lambda = \ln 2 \tag{1.2.11.4}$$

The following code verifies each option and the solution

codes/q11/q11.c

1.2.12 Consider the following Python declarations of two lists.

```
A = [1, 2, 3]
B = [4, 5, 6]
```

Which one of the following statements results in $A = [1, 2, 3, 4, 5, 6]$? (DA 2025)

i) A.extend(B) ii) A.append(B) iii) A.update(B) iv) A.insert(B)

Solution: From (1.1.17.1):

$$A.\text{extend}(B) \implies A \leftarrow [1, 2, 3, 4, 5, 6] \tag{1.2.12.1}$$

The following code verifies the solution

codes/q14/q14.py

1.2.13 Consider two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to (1, \infty)$. Both functions are differentiable at a point c. Which of the following functions is/are ALWAYS differentiable at c? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions. (DA 2025)

i) $f \pm g$ ii) $f \cdot g$ iii) $\frac{f}{g}$ iv) $f \circ g + g \circ f$

Solution: From (1.1.18.1), the product $f \cdot g$, sum $f \pm g$, and quotient f/g (since $g(c) > 1 \neq 0$) are strictly differentiable at c. $f \circ g$ requires differentiability of f at $g(c)$, which is not guaranteed.

1.2.14 Which of the following statements is/are correct? (DA 2025)

i) $\mathbb{R}^n$ has a unique set of orthonormal basis vectors
ii) $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors
iii) Linearly independent vectors in $\mathbb{R}^n$ are orthonormal
iv) Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

Solution: From (1.1.19.1), $\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij}$ guarantees that orthonormal sets are unconditionally linearly independent. Furthermore, orthonormal bases in $\mathbb{R}^n$ are non-unique.

To illustrate these properties with examples:

- Non-uniqueness of Orthonormal Bases in $\mathbb{R}^n$: Consider $\mathbb{R}^2$. The standard canonical orthonormal basis is:

$$\mathbf{B}_1 = \left\{ \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \tag{1.2.14.1}$$

  If we apply a spatial rotation by an angle $\theta = \frac{\pi}{4}$ (45°), we obtain a completely different valid orthonormal basis:

$$\mathbf{B}_2 = \left\{ \mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\} \tag{1.2.14.2}$$

  Since $\mathbf{u}_1^\top \mathbf{u}_2 = 0$ and $\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = 1$, $\mathbf{B}_2$ satisfies all orthonormal basis requirements. This proves that $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors.

- Linearly Independent Vectors Need Not Be Orthogonal/Orthonormal: Consider two vectors in $\mathbb{R}^2$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.14.3}$$

  These vectors are linearly independent because neither is a scalar multiple of the other. However, they are not orthogonal (and thus not orthonormal) since their

inner product is non-zero:

$$\mathbf{v}_1^\top \mathbf{v}_2 = (1)(1) + (0)(1) = 1 \neq 0 \tag{1.2.14.4}$$

This demonstrates that linear independence does not imply orthogonality or orthonormality.

1.2.15 For which of the following inputs does binary search take time $O(\log n)$ in the worst case? (DA 2025)

i) An array of n integers in any order
ii) A linked list of n integers in any order
iii) An array of n integers in increasing order
iv) A linked list of n integers in increasing order

Solution: From (1.1.20.1) and (1.1.20.2):

$$T(n) = T(n/2) + O(1) \tag{1.2.15.1}$$
$$= O(\log n) \tag{1.2.15.2}$$

which requires $O(1)$ random access to the median element, satisfied exclusively by an array of n integers in increasing order.
The following code verifies the solution

codes/q15/q15.c

The following code verifies the theoretical time complexity with simulated values and generates Fig. 1.2.15.1.

codes/q15/q15.py

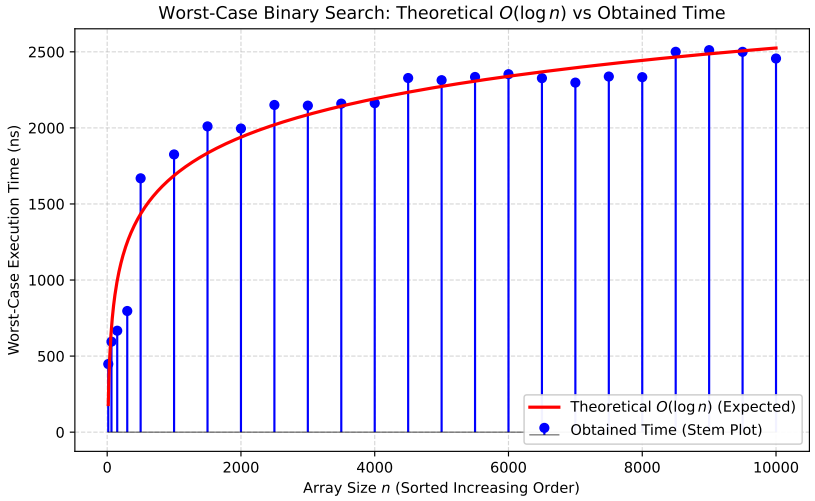


Fig. 1.2.15.1: Binary search time complexity

1.2.16 Let $\mathbf{A} = \mathbf{I}_n + \mathbf{x}\mathbf{x}^\top$ where $\mathbf{I}_n$ is the $n \times n$ identity matrix and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top\mathbf{x} = 1$. Which of the following options is/are correct? (DA 2025)

i) Rank of $\mathbf{A}$ is n
ii) $\mathbf{A}$ is invertible
iii) 0 is an eigenvalue of $\mathbf{A}$
iv) $\mathbf{A}^{-1}$ has a negative eigenvalue

Solution: From (1.1.21.1), evaluating the eigenvalues for $\mathbf{x}^\top\mathbf{x} = 1$:

$$\mathbf{A}\mathbf{x} = \mathbf{x} + \mathbf{x}\left(\mathbf{x}^\top\mathbf{x}\right) \tag{1.2.16.1}$$
$$= 2\mathbf{x} \tag{1.2.16.2}$$
$$\implies \lambda_1 = 2 \tag{1.2.16.3}$$
$$\mathbf{A}\mathbf{v} = \mathbf{v} + \mathbf{x}\left(\mathbf{x}^\top\mathbf{v}\right) \tag{1.2.16.4}$$
$$= 1\mathbf{v} \tag{1.2.16.5}$$
$$\implies \lambda_{2,\ldots,n} = 1 \quad \text{for } \mathbf{x}^\top\mathbf{v} = 0 \tag{1.2.16.6}$$

Since all $\lambda_i > 0$, 0 is not an eigenvalue of $\mathbf{A}$, $\text{rank}\left(\mathbf{A}\right) = n$, and $\mathbf{A}$ is invertible. Furthermore, the eigenvalues of the inverse matrix $\mathbf{A}^{-1}$ are the reciprocals of the eigenvalues of $\mathbf{A}$:

$$\lambda_1\left(\mathbf{A}^{-1}\right) = \frac{1}{\lambda_1} = \frac{1}{2} > 0 \tag{1.2.16.7}$$
$$\lambda_{2,\ldots,n}\left(\mathbf{A}^{-1}\right) = \frac{1}{\lambda_{2,\ldots,n}} = 1 > 0 \tag{1.2.16.8}$$

Because all eigenvalues of $\mathbf{A}^{-1}$ are strictly positive, $\mathbf{A}^{-1}$ has no negative eigenvalues. The following code verifies our solution

codes/q16/q16.c

1.2.17 Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x. (DA 2025)

i) 10 ii) 12 iii) 14 iv) 16

Solution: From (1.1.22.1):

$$\text{Swaps} = \text{Inversions} = 2 \tag{1.2.17.1}$$

The following code finds the number of swaps for the values of x as mentioned in the options

codes/q17/q17.c

1.2.18 There are three boxes containing white balls and black balls.

- Box-1 contains 2 black and 1 white balls.
- Box-2 contains 1 black and 2 white balls.
- Box-3 contains 3 black and 3 white balls.

In a random experiment, one of these boxes is selected, where the probability of choosing Box-1 is $\frac{1}{2}$, Box-2 is $\frac{1}{6}$, and Box-3 is $\frac{1}{3}$. A ball is drawn at random from the selected box. Given that the ball drawn is white, the probability that it is drawn from Box-2 is _____. (DA 2025)

Solution: From (1.1.23.1):

$$Pr\left(E_2 \mid W\right) = \frac{Pr\left(E_2\right) Pr\left(W \mid E_2\right)}{\sum_{i=1}^{3} Pr\left(E_i\right) Pr\left(W \mid E_i\right)} \tag{1.2.18.1}$$

$$= \frac{\frac{1}{6} \cdot \frac{2}{3}}{\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{3}{6}} \tag{1.2.18.2}$$

$$= \frac{\frac{1}{9}}{\frac{1}{6} + \frac{1}{9} + \frac{1}{6}} \tag{1.2.18.3}$$

$$= \frac{\frac{1}{9}}{\frac{4}{9}} \tag{1.2.18.4}$$

$$= \frac{1}{4} = 0.25 \tag{1.2.18.5}$$

1.2.19 $\lim\limits_{t \to +\infty} \sqrt{t^2 + t} - t =$ _____. (DA 2025)

Solution: From (1.1.24.1), with $a = 1$:

$$\lim_{t \to +\infty} \left(\sqrt{t^2 + t} - t\right) = \frac{1}{2} = 0.5 \tag{1.2.19.1}$$

The following code verifies our solution

codes/q19/q19.c

1.2.20 Given data $\{(-1,1),(2,-5),(3,5)\}$ of the form (x,y), we fit a model $y=wx$ using linear least-squares regression. The optimal value of w is _____. (DA 2025)

Solution: From (1.1.27.1), we define the design vector $\mathbf{X}$ and target vector $\mathbf{y}$ for the given data points:

$$\mathbf{X}=\begin{pmatrix}-1\\2\\3\end{pmatrix},\quad \mathbf{y}=\begin{pmatrix}1\\-5\\5\end{pmatrix} \tag{1.2.20.1}$$

The optimal parameter w^* using the matrix least-squares normal equations is given by:

$$w^*=\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y} \tag{1.2.20.2}$$

$$\mathbf{X}^\top\mathbf{X}=\begin{pmatrix}-1 & 2 & 3\end{pmatrix}\begin{pmatrix}-1\\2\\3\end{pmatrix}=14 \tag{1.2.20.3}$$

$$\mathbf{X}^\top\mathbf{y}=\begin{pmatrix}-1 & 2 & 3\end{pmatrix}\begin{pmatrix}1\\-5\\5\end{pmatrix}=4 \tag{1.2.20.4}$$

$$w^*=\frac{4}{14}\approx 0.286 \tag{1.2.20.5}$$

The following code verifies our solution

codes/q20/q20.py

1.2.21 The naive Bayes classifier is used to solve a two-class classification problem with class-labels y_1,y_2. Suppose the prior probabilities are $P(y_1)=\frac{1}{3}$ and $P(y_2)=\frac{2}{3}$. Assuming a discrete feature space with

$$P(\mathbf{x}\mid y_1)=\frac{3}{4}\quad\text{and}\quad P(\mathbf{x}\mid y_2)=\frac{1}{4} \tag{1.2.21.1}$$

for a specific feature vector $\mathbf{x}$. The probability of misclassifying $\mathbf{x}$ is _____. (DA 2025)

Solution: From (1.1.28.1):

$$Pr(\text{error}\mid\mathbf{x})=\frac{\min_k Pr(\mathbf{x}\mid y_k)\,Pr(y_k)}{Pr(\mathbf{x})} \tag{1.2.21.2}$$

$$=\frac{\min\left(\frac{3}{4}\cdot\frac{1}{3},\frac{1}{4}\cdot\frac{2}{3}\right)}{\frac{3}{4}\cdot\frac{1}{3}+\frac{1}{4}\cdot\frac{2}{3}} \tag{1.2.21.3}$$

$$=\frac{\min\left(\frac{1}{4},\frac{1}{6}\right)}{\frac{1}{4}+\frac{1}{6}} \tag{1.2.21.4}$$

$$=\frac{\frac{1}{6}}{\frac{5}{12}} \tag{1.2.21.5}$$

$$=\frac{2}{5}=0.4 \tag{1.2.21.6}$$

1.2.22 Let $Y = Z^2,\ Z = \frac{X-\mu}{\sigma}$, where X is a normal random variable with mean μ and variance σ^2. The variance of Y is (DA 2025)

i) 1 ii) 2 iii) 3 iv) 4

Solution: From (1.1.29.1), with $Z \sim \mathcal{N}(0,1)$:

$$var(Y) = E\left[Z^4\right] - \left(E\left[Z^2\right]\right)^2 \quad \text{(using (1.1.29.2))} \tag{1.2.22.1}$$
$$= 3 - 1^2 \tag{1.2.22.2}$$
$$= 2 \tag{1.2.22.3}$$

The following code verifies our solution

codes/q22/q22.c

1.2.23 Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be such that $\mathbf{A}^3 = \mathbf{A}$. Which one of the following statements is ALWAYS correct? (DA 2025)

i) $\mathbf{A}$ is invertible
ii) Determinant of $\mathbf{A}$ is 0
iii) The sum of the diagonal elements of $\mathbf{A}$ is 1
iv) $\mathbf{A}$ and $\mathbf{A}^2$ have the same rank

Solution: From (1.1.30.1):

$$\mathbf{A}^3 = \mathbf{A} \tag{1.2.23.1}$$
$$\implies \text{rank}(\mathbf{A}) = \text{rank}\left(\mathbf{A}^2\right) \tag{1.2.23.2}$$

To demonstrate that other properties are not universally guaranteed, consider counterexamples satisfying $\mathbf{A}^3 = \mathbf{A}$:

- For the zero matrix $\mathbf{A} = \mathbf{0}$, $\left|\mathbf{A}\right| = 0$ and it is not invertible, disproving strict invertibility and confirming determinant can be 0.
- For the identity matrix $\mathbf{A} = \mathbf{I}_n$, $\left|\mathbf{A}\right| = 1 \neq 0$ and the trace is n (or 2 for $\mathbf{I}_2$), showing that the determinant is not always 0 and the sum of diagonal elements is not universally 1.

1.2.24 Let $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i,j)-th element of matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ be given by $A_{ij} = \mathbf{x}_i^\top \mathbf{x}_j,\ 1 \leq i,j \leq n$. Which one of the following statements is correct? (DA 2025)

i) $\mathbf{A}$ is invertible
ii) 0 is a singular value of $\mathbf{A}$
iii) Determinant of $\mathbf{A}$ is 0
iv) $\mathbf{z}^\top \mathbf{A}\mathbf{z} = 0$ for some non-zero $\mathbf{z} \in \mathbb{R}^n$

Solution: From (1.1.31.5), the following properties hold:

- Positive Definiteness: For every $\mathbf{z} \neq \mathbf{0}$, $\mathbf{X}\mathbf{z} \neq \mathbf{0}$, which ensures $\|\mathbf{X}\mathbf{z}\|^2 > 0$. Consequently:

$$\mathbf{z}^\top \mathbf{A}\mathbf{z} > 0, \quad \forall \mathbf{z} \neq \mathbf{0} \tag{1.2.24.1}$$

which implies $\mathbf{z}^\top \mathbf{A}\mathbf{z} \neq 0$ for any non-zero $\mathbf{z}$.

- Eigenvalues and Invertibility: Let λ be an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \implies \mathbf{v}^\top \mathbf{A}\mathbf{v} = \lambda \|\mathbf{v}\|^2 \tag{1.2.24.2}$$

Since $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\|^2 > 0$. Together with $\mathbf{v}^\top \mathbf{A}\mathbf{v} > 0$, this yields:

$$\lambda = \frac{\mathbf{v}^\top \mathbf{A}\mathbf{v}}{\|\mathbf{v}\|^2} > 0 \tag{1.2.24.3}$$

Because every eigenvalue $\lambda_i > 0$, zero is not an eigenvalue of $\mathbf{A}$. Hence, the null space is trivial, which proves that $\mathbf{A}$ has full rank n and is invertible.

- Singular Values: Because $\mathbf{A}$ is real, symmetric, and positive definite, its singular values coincide directly with its eigenvalues:

$$\sigma_i = \sqrt{\lambda_i \left(\mathbf{A}^\top \mathbf{A}\right)} = \sqrt{\lambda_i^2} = \lambda_i > 0, \quad \forall i \in \{1, \ldots, n\} \tag{1.2.24.4}$$

Because all singular values are strictly positive, 0 cannot be a singular value of $\mathbf{A}$.

1.2.25 Consider the cumulative distribution function (CDF) of a random variable X:

$$F_X(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{4}(x+1)^2 & -1 \leq x \leq 1 \\ 1 & x \geq 1 \end{cases} \tag{1.2.25.1}$$

The value of $P\left(X^2 \leq 0.25\right)$ is (DA 2025)

i) 0.625 ii) 0.25 iii) 0.5 iv) 0.5625

Solution: From (1.1.32.1), with $c = \sqrt{0.25} = 0.5$:

$$P\left(X^2 \leq 0.25\right) = F_X(0.5) - F_X(-0.5) \quad \text{(using (1.1.32.1))} \tag{1.2.25.2}$$

$$= \frac{1}{4}(0.5+1)^2 - \frac{1}{4}(-0.5+1)^2 \quad \text{(using (1.2.25.1))} \tag{1.2.25.3}$$

$$= \frac{2.25 - 0.25}{4} \tag{1.2.25.4}$$

$$= 0.5 \tag{1.2.25.5}$$

Differentiating the cumulative distribution function $F_X(x)$ with respect to x yields the probability density function (PDF), $f_X(x) = \frac{d}{dx} F_X(x)$:

$$f_X(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{2}(x+1) & -1 < x < 1 \\ 0 & x > 1 \end{cases} \tag{1.2.25.6}$$

The following code verifies the solution and generates the data for simulating the CDF

codes/q25/q25.c

The following code verifies the theoretical CDF with simulated values and generates Fig. 1.2.25.1.

codes/q25/q25.py

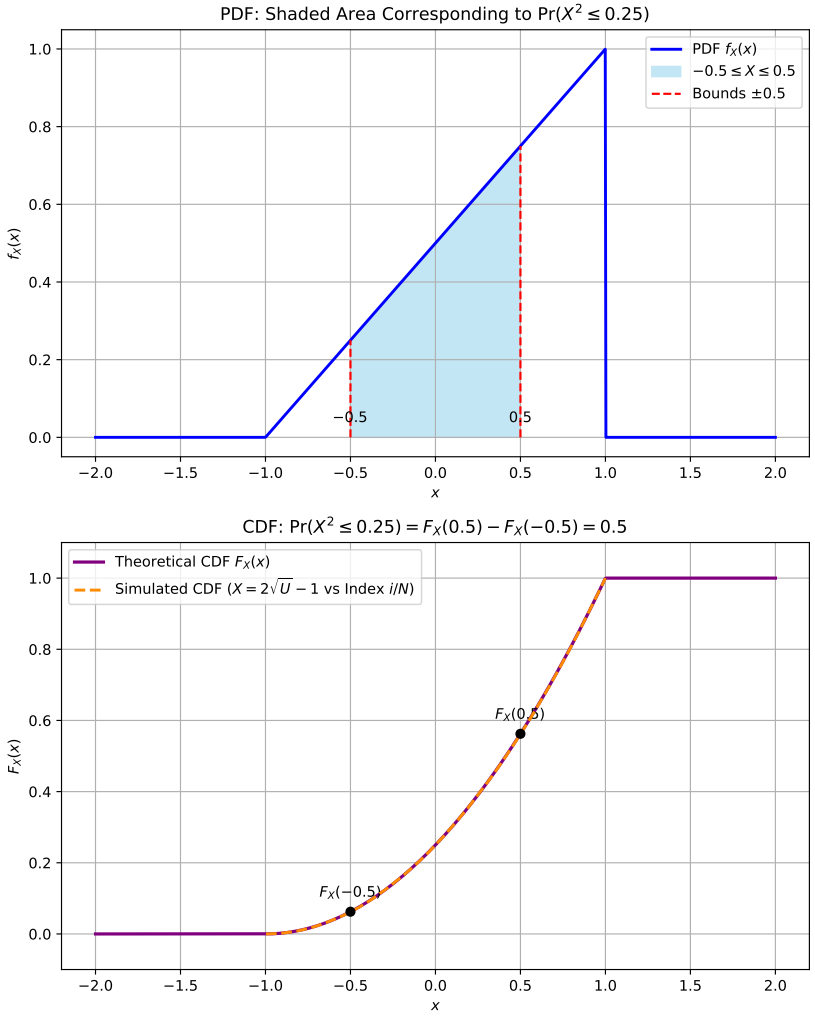


Fig. 1.2.25.1: PDF and CDF of continuous random variable X illustrating $P(X^2 \leq 0.25)$

1.2.26 A random variable X is said to be distributed as Bernoulli(θ), denoted by $X \sim$

Bernoulli(θ), if $P(X=1)=\theta,\ P(X=0)=1-\theta$ for $0<\theta<1$. Let $Y=\sum_{i=1}^{300} X_i$, where $X_i \sim$ Bernoulli(θ), $i=1,2,\ldots,300$ be independent and identically distributed random variables with $\theta=0.25$. The value of $P(60\le Y\le 90)$ after approximation through Central Limit Theorem, is given by (Recall that $\phi(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{t^2}{2}}\,dt$) (DA 2025)

i) $\phi(2)-\phi(-2)$
ii) $\phi(1)-\phi(-1)$
iii) $\phi(3)-\phi(-3)$
iv) $\phi(90)-\phi(60)$

Solution: From (1.1.33.2) and (1.1.33.3), for $n=300$ and $\theta=0.25$:

$$\mu = n\theta = 75 \tag{1.2.26.1}$$

$$\sigma = \sqrt{n\theta(1-\theta)} = 7.5 \tag{1.2.26.2}$$

From (1.1.33.5), with $a=60$ and $b=90$:

$$P(60\le Y\le 90) = \phi\left(\frac{90-75}{7.5}\right) - \phi\left(\frac{60-75}{7.5}\right) \tag{1.2.26.3}$$

$$= \phi(2)-\phi(-2) \tag{1.2.26.4}$$

The following code verifies the Central Limit Theorem approximation for the binomial sum $Y=\sum_{i=1}^{300} X_i$ and generates Fig. 1.2.26.1.

codes/q26/q26.py

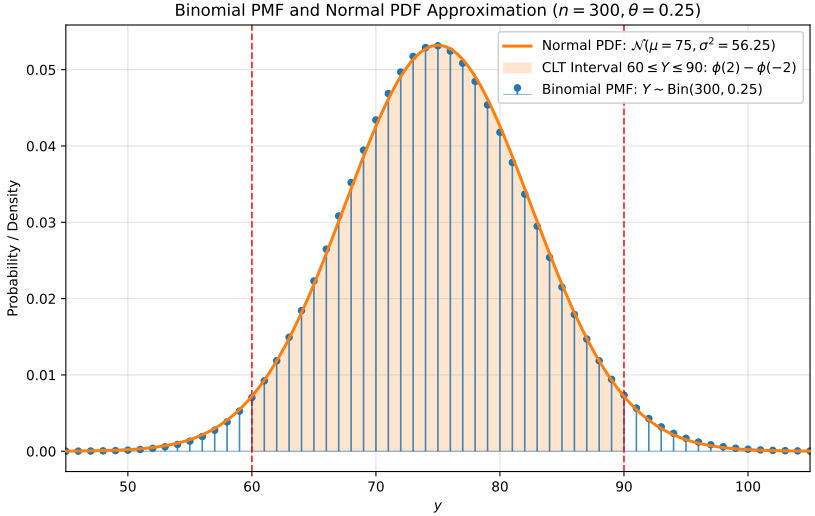


Fig. 1.2.26.1: Binomial PMF and Normal PDF approximation for $Y \sim \text{Bin}(300, 0.25)$ illustrating $P(60\le Y\le 90)\approx\phi(2)-\phi(-2)$

1.2.27 For $x \in \mathbb{R}$, the floor function is denoted by $f(x) = \lfloor x \rfloor$ and defined as follows

$$\lfloor x \rfloor = k, \quad k \le x < k+1 \tag{1.2.27.1}$$

where k is an integer. Let $Y = \lfloor X \rfloor$ where X is an exponentially distributed random variable with mean $\frac{1}{\ln 10}$, where $\ln$ denotes natural logarithm. For any positive integer l, one can write the probability of the event $Y = l$ as follows

$$P(Y = l) = q^l (1 - q) \tag{1.2.27.2}$$

The value of q is (DA 2025)

i) 0.1 ii) 0.01 iii) 0.5 iv) 0.434

Solution: From (1.1.34.1) and (1.1.34.7), with $\lambda = \ln 10$:

$$P(Y = l) = q^l (1 - q) \tag{1.2.27.3}$$
$$q = e^{-\lambda} \tag{1.2.27.4}$$
$$= e^{-\ln 10} \tag{1.2.27.5}$$
$$= 0.1 \tag{1.2.27.6}$$

The following code verifies the solution

codes/q27/q27.c

1.2.28 A random experiment consists of throwing 100 fair dice, each die having six faces numbered 1 to 6. An event A represents the set of all outcomes where at least one of the dice shows a 1. Then, $P(A) =$ (DA 2025)

i) 0 ii) 1 iii) $1 - \left(\frac{5}{6}\right)^{100}$ iv) $\left(\frac{5}{6}\right)^{100}$

Solution: From (1.1.35.4), with $n = 100$ and $q = 5/6$:

$$P(A) = 1 - q^n \tag{1.2.28.1}$$
$$= 1 - \left(\frac{5}{6}\right)^{100} \tag{1.2.28.2}$$

The following code verifies the solution for the dice rolling probability and compares the exact binomial complement formula with the Gaussian (Central Limit Theorem) approximation.

codes/q28/q28.py

1.2.29 Consider the following Python code snippet.

```
A = {"this", "that"}
B = {"that", "other"}
C = {"other", "this"}
```

```
while ”other” in C:
        if ”this” in A:
                A, B, C = A − B, B − C, C − A
        if ”that” in B:
                A, B, C = C | A, A | B, B | C
```

When the above program is executed, at the end, which of the following sets contains `"this"`? (DA 2025)

i) Only A ii) Only B iii) Only C iv) A, C

Solution: The execution proceeds step-by-step as follows:

- Initial state:

$$A_0 = \{\text{‘this’}, \text{‘that’}\}, \quad B_0 = \{\text{‘that’}, \text{‘other’}\}, \quad C_0 = \{\text{‘other’}, \text{‘this’}\} \tag{1.2.29.1}$$

- Iteration 1: (‘other’ $\in C_0$ is true):
  i) Since ‘this’ $\in A_0$, evaluate set differences simultaneously:

$$A_1 = A_0 \setminus B_0 = \{\text{‘this’}\} \tag{1.2.29.2}$$
$$B_1 = B_0 \setminus C_0 = \{\text{‘that’}\} \tag{1.2.29.3}$$
$$C_1 = C_0 \setminus A_0 = \{\text{‘other’}\} \tag{1.2.29.4}$$

  ii) Since ‘that’ $\in B_1$, evaluate set unions simultaneously:

$$A_2 = C_1 \cup A_1 = \{\text{‘other’}, \text{‘this’}\} \tag{1.2.29.5}$$
$$B_2 = A_1 \cup B_1 = \{\text{‘this’}, \text{‘that’}\} \tag{1.2.29.6}$$
$$C_2 = B_1 \cup C_1 = \{\text{‘that’}, \text{‘other’}\} \tag{1.2.29.7}$$

- Iteration 2: (‘other’ $\in C_2$ is true):
  i) Since ‘this’ $\in A_2$, evaluate set differences simultaneously:

$$A_3 = A_2 \setminus B_2 = \{\text{‘other’}\} \tag{1.2.29.8}$$
$$B_3 = B_2 \setminus C_2 = \{\text{‘this’}\} \tag{1.2.29.9}$$
$$C_3 = C_2 \setminus A_2 = \{\text{‘that’}\} \tag{1.2.29.10}$$

  ii) Now check ‘that’ $\in B_3$: since$B_3 = \{\text{‘this’}\}$, this condition is false. The second block does not execute.
- Loop termination: The condition ‘other’ $\in C_3$evaluates to false because$C_3 = \{\text{‘that’}\}$.

At termination, the final sets are:

$$A = \{\text{‘other’}\}, \quad B = \{\text{‘this’}\}, \quad C = \{\text{‘that’}\} \tag{1.2.29.11}$$

$\therefore$ Only set B contains `"this"`.

The following code verifies the solution

codes/q29/q29.py

1.2.30 Consider the function

$$f(x) = \frac{x^3}{3} + \frac{7}{2}x^2 + 10x + \frac{133}{2}, \quad x \in [-8, 0] \tag{1.2.30.1}$$

Which of the following statements is/are correct? (DA 2025)

i) The maximum value of f is attained at $x = -5$
ii) The minimum value of f is attained at $x = -2$
iii) The maximum value of f is $\frac{133}{2}$
iv) The minimum value of the derivative of f is attained at $x = -\frac{7}{2}$

Solution: From (1.1.36.1), setting $f'(x) = 0$:

$$f'(x) = x^2 + 7x + 10 = 0 \tag{1.2.30.2}$$
$$\implies x = -5, \quad x = -2 \tag{1.2.30.3}$$
$$f''(x) = 2x + 7 \tag{1.2.30.4}$$
$$\implies f''(-5) = -3 < 0 \tag{1.2.30.5}$$
$$f''(-2) = 3 > 0 \tag{1.2.30.6}$$

Thus, a local maximum occurs at $x = -5$ and a local minimum occurs at $x = -2$. The minimum of the derivative is found by completing the square:

$$f'(x) = x^2 + 7x + 10 = \left(x + \frac{7}{2}\right)^2 - \frac{9}{4} \tag{1.2.30.7}$$

which attains its minimum at $x = -\frac{7}{2} \in [-8, 0]$.
To find the global extrema on the closed interval $[-8, 0]$, evaluate $f(x)$ at the critical points and the boundary endpoints:

$$f(-8) = \frac{(-8)^3}{3} + \frac{7}{2}(-8)^2 + 10(-8) + \frac{133}{2} = \frac{239}{6} \approx 39.83 \tag{1.2.30.8}$$
$$f(-5) = \frac{(-5)^3}{3} + \frac{7}{2}(-5)^2 + 10(-5) + \frac{133}{2} = \frac{187}{3} \approx 62.33 \tag{1.2.30.9}$$
$$f(-2) = \frac{(-2)^3}{3} + \frac{7}{2}(-2)^2 + 10(-2) + \frac{133}{2} = \frac{347}{6} \approx 57.83 \tag{1.2.30.10}$$
$$f(0) = \frac{(0)^3}{3} + \frac{7}{2}(0)^2 + 10(0) + \frac{133}{2} = \frac{133}{2} = 66.5 \tag{1.2.30.11}$$

Comparing these values, the global maximum on $[-8, 0]$ occurs at the boundary $x = 0$:

$$\max_{x \in [-8,0]} f(x) = f(0) = \frac{133}{2} \tag{1.2.30.12}$$

Because $f(0) = \frac{133}{2} > f(-5) = \frac{187}{3}$, the global maximum is attained at $x = 0$ rather than $x = -5$.

Similarly, the global minimum is $f(-8) = \frac{239}{6}$, attained at the boundary $x = -8$ rather than $x = -2$.

The following code verifies the solution and generates Fig. 1.2.30.1.

codes/q30/q30.py

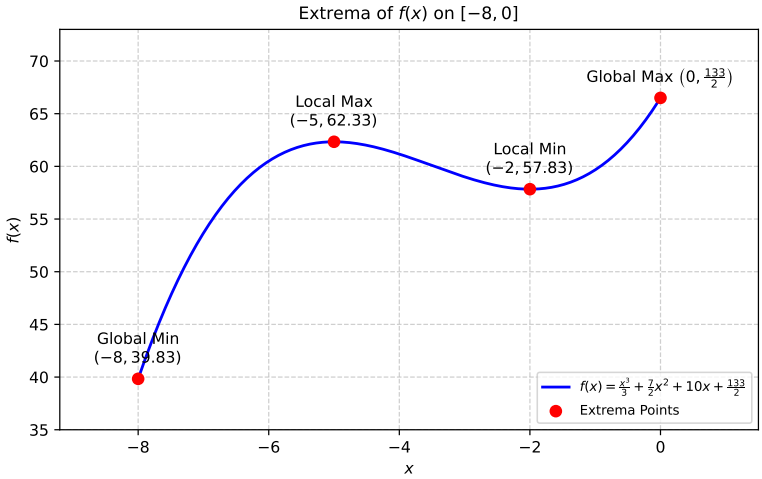


Fig. 1.2.30.1: Extrema of $f(x)$ on $[-8, 0]$

1.2.31 Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $\mathbf{A} = \mathbf{x}_1\mathbf{x}_1^\top + \cdots + \mathbf{x}_5\mathbf{x}_5^\top$. Which of the following statements is/are correct? (DA 2025)

i) Singular values of $\mathbf{A}$ are also its eigenvalues
ii) Singular values of $\mathbf{A}$ are either 0 or 1
iii) Determinant of $\mathbf{A}$ is 1
iv) $\mathbf{A}$ is invertible

Solution: From (1.1.37.1), the orthonormality property $\mathbf{x}_i^\top \mathbf{x}_j = \delta_{ij}$ implies that $\mathbf{A}$ is symmetric and idempotent:

$$\mathbf{A}^\top = \mathbf{A} \tag{1.2.31.1}$$

$$\mathbf{A}^2 = \mathbf{A} \tag{1.2.31.2}$$

Let λ be an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \implies \mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v} \tag{1.2.31.3}$$

Substituting $\mathbf{A}^2 = \mathbf{A}$:

$$\lambda^2\mathbf{v} = \lambda\mathbf{v} \implies \left(\lambda^2 - \lambda\right)\mathbf{v} = \mathbf{0} \tag{1.2.31.4}$$

Since $\mathbf{v} \neq \mathbf{0}$:

$$\lambda(\lambda - 1) = 0 \implies \lambda \in \{0, 1\} \tag{1.2.31.5}$$

Specifically:

- For any $\mathbf{x}_k$ ($k \in \{1, \ldots, 5\}$):

$$\mathbf{A}\mathbf{x}_k = \sum_{i=1}^{5} \mathbf{x}_i \left(\mathbf{x}_i^\top \mathbf{x}_k\right) = \mathbf{x}_k (1) = 1 \cdot \mathbf{x}_k \implies \lambda = 1 \quad \text{(multiplicity 5)} \tag{1.2.31.6}$$

- For any orthogonal vector $\mathbf{y} \perp \text{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_5\}$:

$$\mathbf{A}\mathbf{y} = \sum_{i=1}^{5} \mathbf{x}_i \left(\mathbf{x}_i^\top \mathbf{y}\right) = \mathbf{0} = 0 \cdot \mathbf{y} \implies \lambda = 0 \quad \text{(multiplicity 5)} \tag{1.2.31.7}$$

Since $\mathbf{A}$ is real, symmetric, and positive semi-definite ($\lambda_i \geq 0$), its singular values σ_i are identical to its eigenvalues:

$$\sigma_i = |\lambda_i| = \lambda_i \implies \sigma_i \in \{0, 1\} \tag{1.2.31.8}$$

The determinant is given by the product of all 10 eigenvalues:

$$\left|\mathbf{A}\right| = \prod_{i=1}^{10} \lambda_i = 1^5 \times 0^5 = 0 \tag{1.2.31.9}$$

Since $\left|\mathbf{A}\right| = 0$, $\mathbf{A}$ is non-invertible.
The following code verifies the solution

codes/q31/q31.py

1.2.32 Let $f : \mathbb{R} \to \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct? (DA 2025)

i) f has a local minima
ii) There does not exist x and y, $x \neq y$ such that $f'(x) = f'(y) = 0$
iii) f has at most one global minimum
iv) f has at most one local minimum

Solution: Since $f''(x) > 0$ for all $x \in \mathbb{R}$, the first derivative $f'(x)$ is strictly increasing, and $f(x)$ is strictly convex across $\mathbb{R}$.

- Existence of local minimum: The condition $f''(x) > 0$ guarantees convexity but does not guarantee the existence of a critical point. For instance, $f(x) = e^x$ satisfies $f''(x) = e^x > 0$ for all $x \in \mathbb{R}$, yet $f'(x) = e^x \neq 0$ everywhere, so it has no local or global minima.
- Uniqueness of critical points: Suppose there exist distinct points $x, y \in \mathbb{R}$ with $x < y$ such that $f'(x) = f'(y) = 0$. Applying Rolle's Theorem (1.1.38.1) to the

function $g(t) = f'(t)$ on $[x, y]$:

$$g(x) = g(y) = 0 \implies \exists\, c \in (x, y) \text{ s.t. } g'(c) = f''(c) = 0 \tag{1.2.32.1}$$

This contradicts the given condition that $f''(t) > 0$ for all $t \in \mathbb{R}$. Hence, no such pair of points can exist, meaning $f'(x) = 0$ can occur at at most one point.

- Global and local minima count: For any convex function, any local minimum is necessarily a global minimum. Since $f'(x)$ is strictly increasing, $f'(x) = 0$ has at most one solution, ensuring f has at most one local minimum and at most one global minimum.

1.2.33 An $n \times n$ matrix $\mathbf{A}$ with real entries satisfies the property: $\|\mathbf{Ax}\|^2 = \|\mathbf{x}\|^2$, for all $\mathbf{x} \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct? (DA 2025)

i) $\mathbf{A}$ must be orthogonal
ii) $\mathbf{A} = \mathbf{I}$ where $\mathbf{I}$ denotes the identity matrix, is the only solution
iii) The eigenvalues of $\mathbf{A}$ are either $+1$ or -1
iv) $\mathbf{A}$ has full rank

Solution: From (1.1.39.1):

$$\|\mathbf{Ax}\|^2 = \|\mathbf{x}\|^2 \iff \mathbf{A}^\top \mathbf{A} = \mathbf{I} \tag{1.2.33.1}$$

Thus, $\mathbf{A}$ must be orthogonal, which implies $\left|\mathbf{A}\right| = \pm 1 \neq 0$ and $\text{rank}\,(\mathbf{A}) = n$ (full rank).
However, a real orthogonal matrix need not equal $\mathbf{I}$ and can possess complex eigenvalues of unit modulus $|\lambda| = 1$. For example, the 90° rotation matrix:

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{1.2.33.2}$$

satisfies $\mathbf{A}^\top \mathbf{A} = \mathbf{I}$ with $\mathbf{A} \neq \mathbf{I}$, and its characteristic equation $\lambda^2 + 1 = 0$ yields complex eigenvalues $\lambda = \pm \jmath \notin \{+1, -1\}$.
The following code verifies the solution

codes/q33/q33.py

1.2.34 Consider designing a linear binary classifier $f(\mathbf{x}) = \text{sign}\left(\mathbf{w}^\top \mathbf{x} + b\right)$, $\mathbf{x} \in \mathbb{R}^2$ on the following training data:

$$\text{Class-1} : \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}, \quad \text{Class-2} : \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \tag{1.2.34.1}$$

Hard-margin support vector machine (SVM) formulation is solved to obtain $\mathbf{w}$ and b. Which of the following options is/are correct? (DA 2025)

i) $\mathbf{w} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = 1$
ii) The number of support vectors is 3
iii) The margin is $\sqrt{2}$
iv) Training accuracy is 98%

Solution: Let the training vectors and their corresponding class labels be denoted by:

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \; y_1 = +1; \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \; y_2 = +1 \tag{1.2.34.2}$$

$$\mathbf{x}_3 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \; y_3 = +1; \quad \mathbf{x}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \; y_4 = -1 \tag{1.2.34.3}$$

The nearest points of Class-1 to the origin $\mathbf{x}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ lie along the line segment connecting $\mathbf{x}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$. By definition of support vectors in (1.1.40.3), these active boundary points satisfy $y_i \left(\mathbf{w}^\top \mathbf{x}_i + b \right) = 1$.

For the negative support vector $\mathbf{x}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ with label $y_4 = -1$:

$$-\left(\mathbf{w}^\top \begin{pmatrix} 0 \\ 0 \end{pmatrix} + b \right) = 1 \implies b = -1 \tag{1.2.34.4}$$

For the positive support vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ with labels $y_1 = y_2 = +1$:

$$\mathbf{w}^\top \mathbf{x}_1 + b = 1 \implies \begin{pmatrix} w_1 & w_2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} - 1 = 1 \implies 2w_1 = 2 \implies w_1 = 1 \tag{1.2.34.5}$$

$$\mathbf{w}^\top \mathbf{x}_2 + b = 1 \implies \begin{pmatrix} w_1 & w_2 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} - 1 = 1 \implies 2w_2 = 2 \implies w_2 = 1 \tag{1.2.34.6}$$

Thus, the optimal parameters of the decision boundary are:

$$\mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b = -1 \tag{1.2.34.7}$$

- The solved parameter values are $\mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ and $b = -1 \neq 1$.
- Testing the functional margin $y_i \left(\mathbf{w}^\top \mathbf{x}_i + b \right)$ for all training samples using (1.1.40.3):

$$y_1 \left(\mathbf{w}^\top \mathbf{x}_1 + b \right) = +1 \left(1(2) + 1(0) - 1 \right) = 1 \quad \text{(Support Vector)} \tag{1.2.34.8}$$

$$y_2 \left(\mathbf{w}^\top \mathbf{x}_2 + b \right) = +1 \left(1(0) + 1(2) - 1 \right) = 1 \quad \text{(Support Vector)} \tag{1.2.34.9}$$

$$y_4 \left(\mathbf{w}^\top \mathbf{x}_4 + b \right) = -1 \left(1(0) + 1(0) - 1 \right) = 1 \quad \text{(Support Vector)} \tag{1.2.34.10}$$

$$y_3 \left(\mathbf{w}^\top \mathbf{x}_3 + b \right) = +1 \left(1(2) + 1(2) - 1 \right) = 3 > 1 \tag{1.2.34.11}$$

  Since only $\mathbf{x}_1, \mathbf{x}_2$, and $\mathbf{x}_4$ lie on the margin boundaries $y_i \left(\mathbf{w}^\top \mathbf{x}_i + b \right) = 1$, the number of support vectors is exactly 3.
- The Euclidean norm of $\mathbf{w}$ is $\|\mathbf{w}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$. Substituting $\|\mathbf{w}\|$ into

(1.1.40.4), the geometric margin evaluates to:

$$\gamma = \frac{2}{\|\mathbf{w}\|} = \frac{2}{\sqrt{2}} = \sqrt{2} \tag{1.2.34.12}$$

- All training points strictly satisfy the canonical feasibility condition (1.1.40.2), $y_i\left(\mathbf{w}^\top \mathbf{x}_i + b\right) \geq 1 > 0$. The classifier predicts all 4 out of 4 samples correctly without misclassification:

$$\text{Training Accuracy} = \frac{4}{4} \times 100\% = 100\% \tag{1.2.34.13}$$

The following code verifies the solution and generates Fig. 1.2.34.1.

codes/q34/q34.py

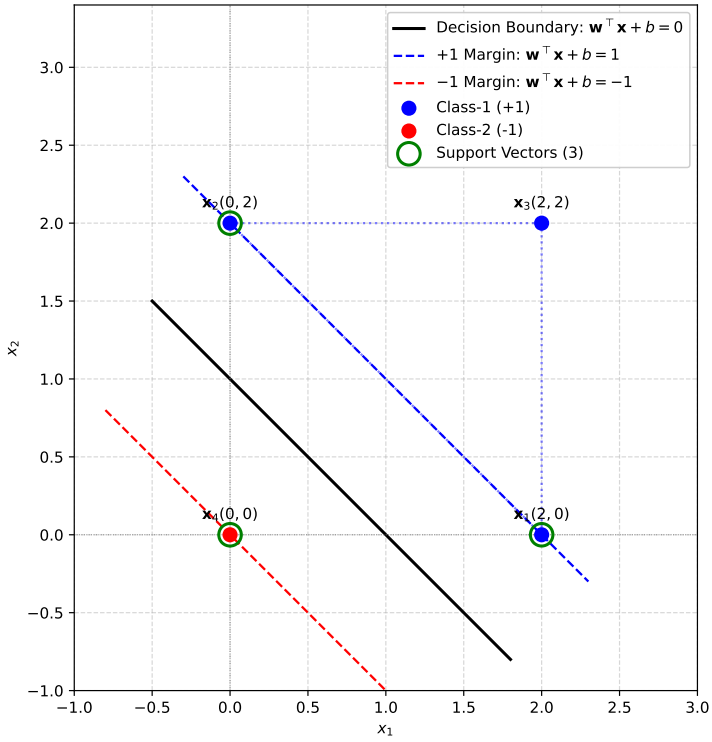


Fig. 1.2.34.1: Hard-Margin SVM decision boundary and margins

1.2.35 Consider a coin-toss experiment where the probability of head showing up is p. In the ith coin toss, let $X_i = 1$ if head appears, and $X_i = 0$ if tail appears. Consider

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{1.2.35.1}$$

where n is the total number of independent coin tosses. Which of the following statements is/are correct? (DA 2025)

i) $E\left[\hat{p}\right] = p$
ii) $E\left[\hat{p}\right] = \frac{p}{n}$
iii) As n increases, variance of $\hat{p}$ decreases
iv) Variance of $\hat{p}$ does not depend on n

Solution: From (1.1.41.1) and (1.1.41.2):

$$E\left[\hat{p}\right] = p \tag{1.2.35.2}$$

$$var\left(\hat{p}\right) = \frac{p\left(1-p\right)}{n} \tag{1.2.35.3}$$

Since n is in the denominator, $var\left(\hat{p}\right)$ strictly decreases as n increases.
The following code verifies the solution

codes/q35/q35.py

1.2.36 Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.

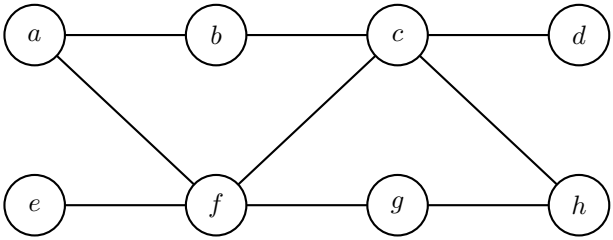


Fig. 1.2.36.1

It is given that $a-b-c-d$ is a shortest path between a and d; $e-f-g-h$ is a shortest path between e and h; $a-f-c-h$ is a shortest path between a and h. Which of the following is/are NOT the edges of G? (DA 2025)

i) (b,d) ii) (b,g) iii) (b,h) iv) (e,g)

Solution: From (1.1.42.1), the shortest path distances given by the three paths of length 3 are $d\left(a,d\right)=3$, $d\left(e,h\right)=3$, and $d\left(a,h\right)=3$:

$$d\left(u,v\right) \le d\left(u,w\right) + d\left(w,v\right) \tag{1.2.36.1}$$

- If edge (b,d) exists, the path $a-b-d$ gives:

$$d\left(a,d\right) \le d\left(a,b\right) + d\left(b,d\right) = 1+1 = 2 < 3 \tag{1.2.36.2}$$

contradicting that $a-b-c-d$ is a shortest path.

- If edge (b,h) exists, the path $a-b-h$ gives:

$$d(a,h) \le d(a,b) + d(b,h) = 1+1 = 2 < 3 \tag{1.2.36.3}$$

  contradicting that $a-f-c-h$ is a shortest path.
- If edge (e,g) exists, the path $e-g-h$ gives:

$$d(e,h) \le d(e,g) + d(g,h) = 1+1 = 2 < 3 \tag{1.2.36.4}$$

  contradicting that $e-f-g-h$ is a shortest path.

$\therefore (b,d)$, (b,h), and (e,g) are not edges of G.
If edge (b,g) exists, the shortest paths using (b,g) between the given pairs are $a-b-g-h$ (length 3), $a-b-c-d$ (length 3), and $e-f-g-h$ (length 3)- None of the shortest-path distances $d(a,d)=3$, $d(e,h)=3$, or $d(a,h)=3$ are reduced below 3.
$\therefore (b,g)$ can be an edge of G.

1.2.37 Let $f:\mathbb{R}\to\mathbb{R}$ be such that $|f(x)-f(y)| \le (x-y)^2$ for all $x,y\in\mathbb{R}$. Then

$$f(1)-f(0) = _____ \tag{1.2.37.1}$$

(DA 2025)

Solution: From (1.1.43.1), with $\alpha = 2 > 1$:

$$|f(x)-f(y)| \le |x-y|^2 \tag{1.2.37.2}$$
$$\implies f'(x) = 0 \tag{1.2.37.3}$$
$$\implies f(x) = c \tag{1.2.37.4}$$
$$\implies f(1)-f(0) = 0 \tag{1.2.37.5}$$

1.2.38 Let $D = \left\{\mathbf{x}^{(1)},\dots,\mathbf{x}^{(n)}\right\}$ be a dataset of n observations where each $\mathbf{x}^{(i)} \in \mathbb{R}^{100}$. It is given that $\sum_{i=1}^{n}\mathbf{x}^{(i)} = \mathbf{0}$. The covariance matrix computed from D has eigenvalues $\lambda_i = 100^{2-i}$, $1 \le i \le 100$. Let $\mathbf{u}\in\mathbb{R}^{100}$ be the direction of maximum variance with $\mathbf{u}^\top\mathbf{u} = 1$. The value of

$$\frac{1}{n}\sum_{i=1}^{n}\left(\mathbf{u}^\top\mathbf{x}^{(i)}\right)^2 = _____ \tag{1.2.38.1}$$

(DA 2025)

Solution: From (1.1.44.1), (1.1.44.2), and (1.1.44.3):

$$\max_{\mathbf{u}^\top\mathbf{u}=1}\frac{1}{n}\sum_{i=1}^{n}\left(\mathbf{u}^\top\mathbf{x}^{(i)}\right)^2 = \max_{\mathbf{u}^\top\mathbf{u}=1}\mathbf{u}^\top\mathbf{C}\mathbf{u} \tag{1.2.38.2}$$
$$= \lambda_{\max} \tag{1.2.38.3}$$

With $\lambda_i = 100^{2-i}$ for $1 \le i \le 100$:

$$\lambda_{\max} = \lambda_1 \tag{1.2.38.4}$$
$$= 100^{2-1} \tag{1.2.38.5}$$
$$= 100 \tag{1.2.38.6}$$

1.2.39 A bag contains 5 white balls and 10 black balls. In a random experiment, n balls are drawn from the bag one at a time with replacement. Let S_n denote the total number of black balls drawn in the experiment. The expectation of S_{100} denoted by $E[S_{100}] =$ _____. (DA 2025)

Solution: From (1.1.45.1), with $p = \frac{10}{15} = \frac{2}{3}$ and $n = 100$:

$$E[S_{100}] = \sum_{i=1}^{100} E[I_i] \tag{1.2.39.1}$$

$$= 100p \tag{1.2.39.2}$$

$$= 100\left(\frac{2}{3}\right) \tag{1.2.39.3}$$

$$= \frac{200}{3} \approx 66.67 \tag{1.2.39.4}$$

1.2.40 Consider the following tables,Loan and Borrower, of a bank.

Loan

loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower

customer_name	loan_num
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Fig. 1.2.40.1: Loan and Borrower relations of a bank

Query: $\pi_{\text{branch_name,customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan})$ where $\bowtie$ denotes natural join. The number of tuples returned by the above relational algebra query is _____. (DA 2025)

Solution: From (1.1.46.1), the divisor relation is the set of all unique branches in `Loan`:

$$\pi_{\text{branch_name}}(\text{Loan}) = \{\text{Banjara Hills}, \text{Kondapur}, \text{SR Nagar}, \text{Balanagar}\} \tag{1.2.40.1}$$

The natural join Loan $\bowtie$ Borrower maps each customer to their respective loan branches:

- Anand: {Banjara Hills} (via L11)
- Ankita: {SR Nagar} (via L15)
- Gopal: {SR Nagar} (via L19)

- Sunil: $\{\text{Balanagar}, \text{Kondapur}\}$ (via L23, L25)
- Karteek: $\{\text{Banjara Hills}, \text{Kondapur}, \text{SR Nagar}, \text{Balanagar}\}$ (via L11, L14, L22, L23)

By definition of relational division $R(A,B) \div S(B)$, the result contains all distinct values of A (`customer_name`) that are paired with every value in S (`branch_name`):

$$\pi_{\text{branch_name,customer_name}}(\text{Loan} \bowtie \text{Borrower}) \div \pi_{\text{branch_name}}(\text{Loan}) = \{\text{Karteek}\} \tag{1.2.40.2}$$

$$\text{Number of tuples} = 1 \tag{1.2.40.3}$$

The following code verifies the solution

codes/q40/q40.c

1.2.41 Consider the following Python code snippet.

```
def f(a, b):
        if (a == 0):
                return b
        if (a % 2 == 1):
                return 2 * f((a - 1) / 2, b)
        return b + f(a - 1, b)

print(f(15, 10))
```

The value printed by the code snippet is _____. (DA 2025)

Solution: The function $f(a,b)$ is defined recursively. Whenever a is odd ($a \pmod 2 = 1$), it branches into $2 \cdot f\left(\frac{a-1}{2}, b\right)$. Unrolling the recursive calls starting with $a = 15$ and $b = 10$:

$$f(15,10) = 2 \cdot f(7,10) \tag{1.2.41.1}$$

$$f(7,10) = 2 \cdot f(3,10) \tag{1.2.41.2}$$

$$f(3,10) = 2 \cdot f(1,10) \tag{1.2.41.3}$$

$$f(1,10) = 2 \cdot f(0,10) \tag{1.2.41.4}$$

When $a = 0$, the base condition triggers and returns $b = 10$:

$$f(0,10) = 10 \tag{1.2.41.5}$$

Substituting the base case back into the recurrence:

$$f(15,10) = 2^4 \cdot f(0,10) = 16 \times 10 = 160 \tag{1.2.41.6}$$

Thus, the value printed by the code snippet is 160.
The following code verifies the solution

codes/q41/q41.py

1.2.42 Consider the following pseudocode.

```
Create empty stack S
Set x = 0, flag = 0, sum = 0
Push x onto S

while (S is not empty) {
        if (flag equals 0){
                Set x = x + 1
                Push x onto S
        }
        if (x equals 8):
                Set flag = 1
                if (flag equals 1){
                        x = Pop(S)
                        if (x is odd):
                        Pop(S)
                        Set sum = sum + x
                }
}
        Output sum
```

The value of sum output by a program executing the above pseudocode is _____. (DA 2025)

Solution: The algorithm executes in two distinct phases:

- Pushing Phase (flag $= 0$): In each iteration, x is incremented and pushed onto stack S. When x reaches 8, the stack contents from bottom to top are:

$$S = [0, 1, 2, 3, 4, 5, 6, 7, 8] \tag{1.2.42.1}$$

  The condition `if (x equals 8)` triggers, setting flag $= 1$.
- Popping Phase (flag $= 1$): Because braces $\{\}$ are not used after `if (x is odd)`:, standard pseudocode scoping rules apply the condition solely to the single next statement `Pop(S)`. The statement `Set sum = sum + x` belongs to the outer `if (flag equals 1)` block and executes on every pop iteration:
  i) Pop $x = 8$ (even): no extra pop occurs $\implies$ sum $= 0 + 8 = 8$.
  ii) Pop $x = 7$ (odd): additionally pops 6 $\implies$ sum $= 8 + 7 = 15$.
  iii) Pop $x = 5$ (odd): additionally pops 4 $\implies$ sum $= 15 + 5 = 20$.
  iv) Pop $x = 3$ (odd): additionally pops 2 $\implies$ sum $= 20 + 3 = 23$.
  v) Pop $x = 1$ (odd): additionally pops 0 $\implies$ sum $= 23 + 1 = 24$.

The stack is now empty, terminating the loop.

$$\text{sum} = 24 \tag{1.2.42.2}$$

The following code verifies the solution

codes/q42/q42.py

1.2.43 Consider a directed graph $G = (V, E)$, where $V = \{0, 1, 2, \ldots, 100\}$ and $E = \{(i, j) : 0 < j - i \leq 2 \text{ for all } i, j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 0. The number of vertices that will be discovered after vertex 50 is _____. (DA 2025)

Solution: From (1.1.47.2), for $0 \leq k \leq 50$:

$$D(k) = 2k \tag{1.2.43.1}$$

$$\implies 2k = 50 \tag{1.2.43.2}$$

$$\implies k_{50} = 25 \tag{1.2.43.3}$$

From (1.1.47.3), the remaining discoveries evaluate to:

$$n - k_{50} = 100 - 25 \tag{1.2.43.4}$$

$$= 75 \tag{1.2.43.5}$$